

CORE MATHEMATICS GRADE 10: ROTATION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.3 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.3: Rotation
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Properties of rotation (centre, angle, direction — clockwise and anticlockwise) – Rotation on a plane surface (using construction to locate images given centre and angle of rotation) – Rotation on the Cartesian plane (rotating objects given the centre and angle of rotation using coordinates) – Determining the centre and angle of rotation given an object and its image (construction method) – Rotational symmetry of plane figures (locating points of symmetry; determining order of rotational symmetry) – Axis and order of rotational symmetry in solids (pyramids, triangular prisms, cones, tetrahedrons) – Rotation and congruence (deducing congruence — opposite congruence — from rotation) – Application of rotation in real-life situations
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) determine the properties of rotation in different situations, b) rotate an object given the centre and angle of rotation on a plane surface and the Cartesian plane, c) determine the centre and angle of rotation given an object and its image, d) determine the order of rotational symmetry of plane figures, e) determine the axis and order of rotational symmetry in solids, f) deduce congruence from rotation, g) appreciate the application of rotation in real-life situations.
Core Competencies	– Critical Thinking and Problem Solving: the learner draws objects and their images and through construction establishes the centres and angles of rotation, analysing spatial relationships on both plane surfaces and the Cartesian plane. – Communication and Collaboration: the learner discusses using an object and its image on a plane surface to generate the properties of rotation, sharing reasoning with peers during group investigations involving clock hands, paper cutouts, and solid figures collected from the environment.
Core Values	– Responsibility: the learner takes care of learning resources collected from the immediate environment (e.g., pyramids, cones, triangular prisms) to establish their axis and order of rotational symmetry. – Respect: learners appreciate each other's opinion as they discuss and draw the image under a rotation given the object, centre and angle of rotation.
Science & Engineering	– Developing and Using Models: learners construct geometric models on plane surfaces and the Cartesian plane to represent

Practices	and predict the position of rotated images, revising their models as they discover properties of rotation. – Planning and Carrying Out Investigations: learners carry out structured investigations using improvised clocks, paper cutouts, and collected solids to generate and test the properties of rotational symmetry. – Analysing and Interpreting Data: learners analyse coordinate pairs of objects and their images to determine patterns (e.g., invariant centre, preserved distances), deduce the angle of rotation, and identify congruence relationships.
Pertinent & Contemporary Issues (PCIs)	– Environmental Education: as the learner responsibly collects regular solids (pyramids, cones, tetrahedrons, triangular prisms) from the immediate environment to determine the axis and order of rotational symmetry, promoting mindful interaction with natural and manufactured objects.
Career Connections	– Architecture and Engineering: rotation is fundamental in the design of buildings, bridges, and mechanical components — e.g., the rotating turbines at Olkaria Geothermal Plant in the Rift Valley rely on rotational symmetry for efficient energy generation. – Graphic Design and Animation: rotation transformations underpin digital design, logo creation, and animation used in Kenya's growing creative industry, including media houses in Nairobi. – Manufacturing and Textile Design: rotational patterns are used in fabric and kitenge print design, as well as in the machining of gear-driven equipment in Kenyan agro-processing factories. – Aviation and Navigation: pilots and navigators use angular rotation concepts when plotting flight paths and bearing changes, relevant to Kenya's busy airspace managed through Wilson Airport and JKIA in Nairobi. – Medicine and Medical Imaging: CT scanners and MRI machines use rotational mechanics; understanding rotation supports careers in biomedical engineering and radiography at hospitals such as Kenyatta National Hospital.
Focus for Lessons	This unit develops learners' understanding of rotation as a geometric transformation, progressing from hands-on exploration of the properties of rotation using improvised clocks and paper cutouts, to precise construction on plane surfaces and the Cartesian plane, and finally to the identification of rotational symmetry in both plane figures and three-dimensional solids. The Kenyan context is woven throughout — from the rotating hands of the clock tower at the University of Nairobi, to the rotational symmetry of Maasai shields and the gear systems in Nairobi's industrial area — grounding abstract mathematics in familiar, observable phenomena. Learners also deduce congruence from rotation, connecting this sub-strand to the broader theme of geometric transformations in Strand 2.0.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: When the minute hand of the clock at the University of Nairobi completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up — and how can we use those rules to rotate any shape to any position, on paper and on a grid? KICD KEY INQUIRY QUESTION: 1. How is rotation applied in real-life situations?
Anchoring Phenomenon	THE ANCHORING PHENOMENON — "THE SPINNING HANDS OF NAIROBI'S CLOCK TOWER" Students are shown a short video clip (or a large printed image sequence) of the clock on the Administration Block of the University of Nairobi. The clip captures the minute hand at four distinct positions over one hour: 12:00, 12:15, 12:30, and 12:45. The hand is clearly the same length and shape at every position, yet it points in a completely different direction each time. Students are asked to observe and discuss: The minute hand has moved — but what exactly has changed, and what has stayed the same? The tip of the hand traces a perfect circular arc. The hand itself does not stretch, shrink, or flip. Every point on the hand remains exactly the same distance from the centre of the clock face. Yet the hand has unmistakably moved.

	This is puzzling because it challenges learners to articulate what kind of motion this is — it is neither a slide (translation) nor a mirror flip (reflection). Something is turning about a fixed point. The questions that naturally arise are: Where is the fixed point? How much has the hand turned? Could we predict exactly where the hand will be after any given amount of time? And — if we placed the hand on a coordinate grid, could we calculate the new position of every point on it mathematically? This phenomenon anchors the entire unit: every lesson returns to the idea of a fixed centre, a measurable angle, a preserved shape and size, and a predictable new position — the four pillars of rotation as a geometric transformation.
Storyline Thread	Lesson 1 — PREDICT: "What is happening to the clock hand?" Learners observe the anchoring phenomenon (clock tower video/image sequence). In groups they predict what has changed and what has stayed the same. They sketch the hand at each position and attempt to describe the motion in their own words. The teacher records predictions on the Driving Question Board (DQB). Key vocabulary — centre, angle, direction, clockwise, anticlockwise — is introduced through discussion. Learners use an improvised clock (a card arrow pinned to a board) to demonstrate and discover that distance from centre is preserved. Lesson 2 — OBSERVE & EXPLORE: "Discovering the Properties of Rotation" Learners use paper cutouts of plane figures (triangles, quadrilaterals) and a pin (as the centre of rotation) to physically rotate objects through specified angles on plain paper. They record observations: object and image are congruent; every point moves through the same angle; the centre is the only invariant point; direction matters. Groups share findings and a class list of Properties of Rotation is compiled and added to the DQB. Connection to the posho mill supporting phenomenon is made. Lesson 3 — EXPLAIN & PRACTISE: "Rotating Objects on a Plane Surface" Learners use a geometrical set (ruler, compass, protractor) to construct the image of a given object under a specified rotation (centre and angle given) on plain paper. Step-by-step construction method is developed collaboratively: locate centre, draw rays from centre through each vertex, measure the angle, mark the image vertex at equal distance. Multiple examples with different centres (inside, outside, on the object) and angles (90°, 180°, 270°) are practised. Learners connect this to the clock-hand phenomenon — the centre of the clock is always fixed. Lesson 4 — EXPLAIN & PRACTISE: "Rotation on the Cartesian Plane" Learners transfer the construction skill to the Cartesian plane. Using coordinate grids, they rotate objects about the origin and about other specified centres through 90°, 180°, and 270° (clockwise and anticlockwise). They discover and record coordinate rules (e.g., 90° anticlockwise about the origin: $(x, y) \rightarrow (-y, x)$) through guided pattern-spotting. The Kasarani football rolling phenomenon is revisited — learners calculate where a marked point would appear after a 90° and 180° rotation. Lesson 5 — INVESTIGATE: "Finding the Centre and Angle of Rotation" Given only an object and its image (no centre or angle provided), learners use perpendicular bisector construction to locate the centre of rotation and a protractor to measure the angle. Group investigation on plain paper and Cartesian plane. Results are verified by rotating the object using the found centre and angle and checking it maps onto the image. Learners return to the DQB and update: "We can work backwards — find the centre and angle from the evidence." Lesson 6 — EXPLORE: "Rotational Symmetry of Plane Figures and Solids" Learners use paper cutouts of regular polygons (equilateral triangle, square, regular hexagon — connecting to patterns on a Maasai kiondo basket) to locate points of rotational symmetry and determine the order of rotational symmetry for each figure. They then collect or examine models of 3-D solids (pyramids, triangular prisms, cones, tetrahedrons — connecting to the Rift Valley windmill phenomenon) and identify the axis and order of rotational symmetry for each solid. Findings are recorded in a class table on the DQB.

	Lesson 7 — EXPLAIN: "Rotation and Congruence" Learners use pairs of object-and-image cutouts from previous lessons to classify the type of congruence produced by rotation (opposite congruence — the image cannot be superimposed without lifting from the plane, unlike direct congruence from translation). They compare this with congruence from reflection (also opposite) and from translation (direct). A sorting activity consolidates understanding. Learners connect to the clock: the hour hand at 3:00 and at 9:00 are congruent — but one is a rotation of the other, not a reflection. Lesson 8 — MODEL BUILDING, APPLICATION & ASSESSMENT: "Putting It All Together" Learners revisit the anchoring phenomenon: given the clock hand position at 12:00 on a coordinate grid, they calculate its exact position at 12:15, 12:30, 12:45, and 1:00 using rotation rules. They then complete a structured class activity covering all seven SLOs: rotate an object on the Cartesian plane, find a centre and angle of rotation, determine rotational symmetry order of a given figure and solid, and identify congruence from rotation. Final DQB review: learners articulate a full answer to the driving question in writing. Real-life appreciation task: identify two examples of rotation in their school or home environment in Nairobi/their locality and describe the centre, angle, and order of symmetry.
Supporting Phenomena	– The rotating blades of a maize grinding mill (posho mill) in a Kisumu market: learners observe that each blade tip traces a circle around a fixed axle — prompting questions about the centre of rotation, the angle swept per second, and the symmetry of the blade arrangement. – Rotational patterns on a traditional Kikuyu/Maasai woven basket (kiondo): learners examine how the same motif repeats at equal angular intervals around a central point, introducing the concept of order of rotational symmetry in a culturally familiar object. – A football (soccer ball) rolling across the pitch at Kasarani Stadium in Nairobi: learners notice that as the ball rolls, a mark on its surface traces a curved path — the mark rotates about the ball's moving centre, connecting rotation to real-world circular motion. – The rotating spray arm inside a water-pumping windmill on a Rift Valley farm: learners observe that the arm returns to its original orientation after a full 360° turn, and that opposite arms are congruent images of each other — linking rotation to both symmetry and congruence.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (80 minutes): The Spinning Hands of Nairobi's Clock Tower: What Is Really Happening?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon and establish the four pillars of rotation — fixed centre, measurable angle, preserved shape and size, and predictable new position — through guided observation, prediction, and hands-on discovery using an improvised clock.
Knowledge	By the end of the lesson, the learner should be able to: (a) determine the properties of rotation in different situations.
Skills	Critical thinking and problem solving: the learner draws objects and their images and through construction establishes the centres and angles of rotation. Communication and Collaboration: the learner discusses using an object and its image on a plane surface to generate the properties of rotation.
Attitudes	Responsibility: the learner takes care of learning resources collected from the immediate environment to establish their axis and order of rotational symmetry. Respect: learners appreciate each other's opinion as they discuss and draw the image under a rotation given the object, centre and angle of rotation.
Key Inquiry Question	How is rotation applied in real-life situations?
Purpose in Storyline	This is Lesson 1 of the Rotation unit. It launches the anchoring phenomenon — the spinning minute hand of the University of Nairobi Administration Block clock tower. Learners observe, predict, and begin to articulate what kind of motion is at work. They record initial questions on the Driving Question Board (DQB) and use an improvised card-arrow clock to discover that distance from the centre is preserved under rotation. Every subsequent lesson in this unit will return to this phenomenon.
Safety Notes	When learners use pins or thumbtacks to secure the card arrow to the board, the teacher should supervise carefully to prevent injury. Pins should be collected and stored immediately after the activity. No sharp geometrical instruments are required in this lesson.

B. LESSON OVERVIEW

Learners are introduced to the anchoring phenomenon: a short video clip or printed image sequence showing the minute hand of the clock on the University of Nairobi Administration Block at four positions — 12:00, 12:15, 12:30, and 12:45. Working in groups, they observe and predict what has changed and what has stayed the same, sketching the hand at each position and attempting to describe the motion in their own words. The teacher facilitates whole-class discussion, introduces key vocabulary (centre, angle, direction, clockwise, anticlockwise), and records all student questions and predictions on the class Driving Question Board (DQB). Learners then use an improvised clock — a card arrow pinned to a board — to physically demonstrate the motion and discover that the distance of every point on the hand from the centre pin remains constant regardless of how far the hand has turned. The lesson closes with a model-building prompt in which learners sketch the clock hand on plain paper and annotate what they now know.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Identifying	Before viewing the phenomenon, learners are shown only the first image: the minute hand of the	Distribute the blank clock-face worksheet (a printed circle with a centre dot and the numeral 12 at	PREDICT-BEFORE-OBSERVE (Science and Engineering Practice: Asking Questions and	Observable indicator: Every group has produced a sketch with at least three predicted hand

transformations Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Rigid transformations: preserved properties Source: Khan Academy (English - US curriculum) Search ARES for similar readings	University of Nairobi clock pointing straight up at 12:00. In groups of four, learners discuss and write predictions in response to two prompts written on the board: (1) 'After 15 minutes, 30 minutes, and 45 minutes, where do you predict the tip of the hand will be? Sketch your prediction.' (2) 'What do you think will stay exactly the same about the hand as it moves?' Each group sketches three predicted positions of the hand on the blank clock-face worksheet provided. Groups nominate a spokesperson to share one prediction each. Learners do NOT yet see the remaining images — this preserves the cognitive tension of the predict phase.	the top — no other markings). Say exactly: 'You are looking at the clock on the Administration Block of the University of Nairobi. It is 12:00. The minute hand is pointing straight up. I want you — in your groups — to predict: after 15 minutes, after 30 minutes, and after 45 minutes, where will the tip of the hand be? Sketch your prediction on the worksheet. You have 5 minutes.' [START TIMER — 5 minutes group work.] Circulate and listen. Do NOT confirm or correct predictions. After 5 minutes: 'Let's hear one prediction from each group. I will write them on the board. All predictions are valid right now — we are scientists making a hypothesis.' Cold-call 4 groups in turn. Record each group's prediction on the board using their own words. Ask: 'Group 3 said the hand will be pointing to the right after 30 minutes. Does anyone predict something different?' [WAIT 12 seconds.] Cold-call 1–2 dissenters. Then say: 'We are also predicting what stays the same. Group by group — what did you say stays the same?' Cold-call all 6 groups, spending no more than 20 seconds per group. Record key words: length, shape, colour, the centre point. Say: 'Hold these predictions. We are about to look at the evidence.'	Defining Problems). By committing predictions to paper before viewing the phenomenon, learners activate prior knowledge of movement and geometry. The act of sketching forces them to confront what they actually understand about positional change, creating productive cognitive dissonance when the real images are revealed. Recording all predictions publicly on the board signals that every idea is valued and will be revisited — establishing the DQB culture from the first lesson.	positions on the clock face. Teacher scans worksheets during circulation. Listen for: (a) groups who predict the hand 'slides' or 'flips' — these are key misconceptions to surface in Phase 2. (b) groups who spontaneously mention 'the centre stays fixed' — note these for cold-calling during Phase 3. If fewer than half the groups produce a sketch in 5 minutes, extend by 2 minutes and prompt: 'Just draw an arrow on the clock face — where do you think the tip ends up?'
Observe Phase VIDEO: Defining transformations Source: Khan Academy (English - US curriculum) Search ARES for similar videos	The teacher displays the anchoring phenomenon: a sequence of four large printed images (or a short video clip, 60–90 seconds) showing the University of Nairobi Administration Block clock at 12:00, 12:15, 12:30, and 12:45. Each image is shown for	Before playing/displaying the images, say: 'Watch silently the first time. Your only job is to observe. After the first pass, I will play it again and you will fill in the I-Notice / I-Wonder table on your worksheet.' [Display or play the image sequence — first pass, silent.] After first pass: 'Now we go	I NOTICE / I WONDER TABLE (Science and Engineering Practice: Analysing and Interpreting Data). This two-column structure separates observation (evidence) from inference (questions), training learners to distinguish what they can directly see from what they	Observable indicator: Every learner has at least two entries in the I-Notice column and at least one in the I-Wonder column. Every group has posted a sticky note on the DQB. Teacher reads sticky notes as they are posted — sort mentally into: (a) questions about angle/direction (will be addressed

 EXERCISE:

[Determine rotations](#)

Source: Khan Academy (English - US curriculum)
 [Search ARES for similar readings](#)

approximately 30 seconds before the next appears. Learners watch silently for the first full pass, then the sequence is repeated. On the second pass, learners individually complete the 'I Notice / I Wonder' two-column table on the back of their worksheet. After the second pass, groups share and consolidate their 'Notice' and 'Wonder' statements. Each group writes their single most important 'Wonder' question on a sticky note and places it on the class Driving Question Board (DQB), which is displayed prominently on the wall for the entire unit.

through again. As you watch, write in the I-Notice column: what do you observe that is definitely true? In the I-Wonder column: what questions does this raise for you?' [Second pass — 2 minutes, individual writing.] After second pass: 'Share in your groups. You have 2 minutes. Then agree on the ONE question your group most wants answered. Write it on your sticky note.' [2 minutes group discussion.] As groups write sticky notes, move to the DQB and say: 'This board is our Driving Question Board. Every question you place here is a real mathematical question. As we work through this unit, we will come back and answer them one by one.' Invite groups one at a time to place their sticky note and read it aloud. After all groups have posted, read the driving question aloud: 'When the minute hand of the clock at the University of Nairobi completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up — and how can we use those rules to rotate any shape to any position, on paper and on a grid?' Say: 'This is the big question that drives our entire unit. Your sticky notes are steps towards answering it.' Point to 2–3 sticky notes that directly connect to the driving question and name the connection explicitly. Then ask: 'Looking at the four images — what stayed the same about the hand every time?' [WAIT 15 seconds.] Cold-call 3 learners. Expected answers: the length stayed the same / the hand did not get bigger or smaller / the centre did not move. Write these verbatim on the board under the heading

need to explain. The DQB sticky-note ritual is a Phenomenon-Anchored Unit Launch strategy: it externalises student thinking, makes all questions public and valued, and creates a visible record of growing understanding across the unit. Connecting student sticky notes to the driving question shows learners that their own curiosity is mathematically legitimate.

Lessons 2–3), (b) questions about the centre point (Lesson 4), (c) questions about coordinate grids (Lessons 5–6). If a group's sticky note is a statement rather than a question (e.g. 'The hand moves'), prompt gently: 'Can you turn that into a question — what do you want to know about that?'

		'WHAT STAYS THE SAME'. Then ask: 'What changed?' [WAIT 12 seconds.] Cold-call 3 different learners. Expected: the direction / the angle / which way it is pointing. Write under 'WHAT CHANGED'.		
Explain Phase  VIDEO: Determining rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Translations, Rotations, and Reflections (Flexbook) Source: CK-12  Search ARES for similar readings	Each group receives an 'improvised clock kit': a card arrow (20 cm long, clearly marked with a dot at the tip and a dot at the base), a pin or thumbtack, and a sheet of A3 plain paper. Learners pin the base of the card arrow to the centre of the A3 paper. One learner holds the pin steady (the 'centre of the clock') while a second learner rotates the arrow. The group investigates: (1) Mark the tip position at the start. Rotate the arrow a little. Mark the new tip position. Measure the distance from the centre pin to the tip — before and after. (2) Mark the position of a point halfway along the arrow. Rotate again. Measure the distance from centre to that midpoint — before and after. (3) Try rotating clockwise and anticlockwise. Record all measurements in a table: 'Point on the hand Distance from centre BEFORE rotation Distance from centre AFTER rotation'. Groups write a group conclusion statement: 'When the arrow rotates, the distance from the _____ to any point on the arrow _____.' Key vocabulary is introduced here: the teacher circulates and introduces 'centre of rotation', 'angle of rotation', 'clockwise', 'anticlockwise' as they appear naturally in group work.	Distribute the improvised clock kits to each group. Say: 'We are going to test something. Your group has a card arrow, a pin, and paper. Pin the base of the arrow to the centre of the paper — one person holds the pin steady throughout. That pin represents the centre of the clock at the University of Nairobi. Now, before you rotate the arrow, mark the tip with a pencil dot. Measure the distance from the pin to the tip. Write it down.' [Allow 2 minutes.] 'Now rotate the arrow — any amount you like. Mark the new tip position. Measure the distance from the pin to the new tip position. What do you find?' [WAIT 15 seconds.] Cold-call 2 groups immediately: 'Group 2 — what distance did you get before? What distance after?' After both groups respond, ask: 'Is anyone getting a different result?' [WAIT 10 seconds.] If no dissenter, say: 'So across all groups, the distance from the centre to the tip is _____ after rotation. Why does this matter? Think about the Nairobi clock.' [WAIT 12 seconds.] Cold-call 1 learner. Now say: 'Now mark a point halfway along your arrow. Do the same test — measure before and after rotation. What happens to THAT distance?' [Allow 2 minutes group work, then cold-call 3 groups.] After groups report, introduce vocabulary explicitly: 'The pin is called the CENTRE OF ROTATION. The amount the arrow has turned is	HANDS-ON INVESTIGATION WITH STRUCTURED DATA TABLE (Science and Engineering Practice: Planning and Carrying Out Investigations; Using Mathematics). The physical manipulation of the card arrow makes the abstract property of rotation tangible. By measuring before and after, learners generate their own evidence rather than receiving the property as a definition. The structured data table scaffolds systematic recording. Introducing vocabulary after learners have experienced the concept (not before) ensures that technical language is attached to real understanding rather than empty memorisation. Connecting every finding back to the Nairobi clock maintains phenomenon coherence.	Observable indicator: Each group's data table shows at least two measurements before and after rotation, with values matching to within ± 2 mm. Each group has written a conclusion sentence. Teacher checks 3 groups' tables during circulation. Listen for: (a) groups who say 'the distance changed a little' — address measurement error explicitly: 'Let's re-measure carefully. Is the difference bigger than 2 mm?' (b) groups who are confusing distance from centre with the arc length traced by the tip — ask: 'We are measuring a straight line from the pin to the point — not along the curve. Can you show me which measurement that is?'

		called the ANGLE OF ROTATION. If it turns the same way as a clock, we say CLOCKWISE. If it turns the opposite way, we say ANTICLOCKWISE.' Write each term on the board with a diagram. Ask: 'On the Nairobi clock, does the minute hand move clockwise or anticlockwise?' [WAIT 10 seconds.] Cold-call 2 learners. Then ask: 'So what is the one thing that is ALWAYS preserved — always stays the same — when something rotates?' [WAIT 15 seconds.] Cold-call 4 learners. Lead the class to articulate: 'The distance from the centre of rotation to any point on the object stays the same.' Write this as a boxed statement on the board: 'PROPERTY OF ROTATION: Every point on a rotating object stays the same distance from the centre of rotation.'		
Driving Question Board (DQB) Creation  VIDEO: Working with matrices as transformations of the plane Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Translations, Rotations, and Reflections Practice Source: CK-12  Search ARES for similar readings	The teacher returns the class's attention to the DQB. Learners review the sticky notes they posted in Phase 2. They are given two new tasks: (1) As a group, look at your sticky note question. Can you now partially answer it — or has today's lesson raised an even more specific question? Write an 'update note' in a different colour and place it next to the original. (2) The teacher leads the class in adding four permanent 'anchor questions' to the DQB — one for each of the four pillars of rotation introduced today — which will remain throughout the unit. Learners copy these into their exercise books as the unit's guiding questions.	Direct learners' attention back to the DQB. Say: 'At the start of this lesson you posted questions on this board. In the last 20 minutes you have discovered something important. Look at your group's question. Can you now write a partial answer, or a more specific follow-up question, on a new sticky note?' Distribute sticky notes in a second colour. Allow 3 minutes for groups to write update notes and post them. Then say: 'Let me now add four anchor questions that will guide our entire unit — every lesson will answer part of one of these.' Write (or post pre-printed strips) on the DQB: (1) 'Where is the fixed centre — and how does it control the rotation?' (2) 'How much has the object turned — and how do we measure that angle?' (3) 'What is preserved — what	DQB UPDATE PROTOCOL (Science and Engineering Practice: Constructing Explanations). Returning to the DQB mid-unit and at lesson close is a deliberate metacognitive routine that makes learning visible. The two-colour sticky note system (original question + update) externalises knowledge growth: learners can literally see their understanding developing lesson by lesson. The four anchor pillar questions provide a coherent organising framework that prevents the unit from feeling like disconnected skill drills — every lesson is framed as answering part of a bigger question rooted in the Nairobi clock phenomenon.	Observable indicator: Every group has posted an update sticky note. The update should show either (a) a partial answer referencing today's finding ('The distance stays the same because...') or (b) a more precise follow-up question ('How do we measure the angle exactly?'). Teacher reads update notes and identifies 2–3 to highlight at the start of Lesson 2. If a group's update note is identical to their original question, prompt: 'What did you find out today about the arrow and the pin? Can that help you answer even part of your question?'

		stays exactly the same after rotation?' (4) 'Can we predict and calculate the exact new position of every point?' Say: 'These four questions are the four pillars of this unit. We answered Pillar 3 today — distance from the centre is preserved. By the end of this unit, every pillar will be answered.' Circle Pillar 3 and write next to it: 'ANSWERED TODAY — Lesson 1.' Ask: 'Which pillar do you most want to answer next?' [WAIT 12 seconds.] Cold-call 3 learners and note their responses — use these to frame Lesson 2's opening hook.		
Model Building Phase  VIDEO: Examples: Determine Angles of Rotation Source: CK-12  Search ARES for similar videos  EXERCISE: Determine rotations (basic) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	As a closing individual task, each learner sketches the University of Nairobi clock hand at four positions (12:00, 12:15, 12:30, 12:45) on a fresh blank clock face. They annotate their sketch with: (a) a label for the centre of rotation, (b) arrows showing the direction of rotation (clockwise), (c) a note stating the property they discovered: 'Every point stays the same distance from the centre.' Below the sketch, each learner writes a 'Model Statement' — one or two sentences answering: 'What kind of motion is the clock hand performing, and what makes it different from a slide or a flip?' This sketch and statement become Page 1 of the learner's Rotation Model Journal, which they will add to and revise in every subsequent lesson.	Distribute a clean blank clock-face sheet (or direct learners to a fresh page in their exercise books). Say: 'This is the start of your Rotation Model Journal. On this page, sketch the Nairobi clock hand at 12:00, 12:15, 12:30 and 12:45. Label the centre of rotation. Show the direction of rotation with an arrow. Then write your Model Statement: What kind of motion is this? What makes it different from a translation — a slide — or a reflection — a flip?' Allow 7 minutes individual work. Circulate and look at 6–8 learners' sketches. After 7 minutes, cold-call 3 learners to read their Model Statement aloud. After the third reading, synthesise: 'You have all described something important. The hand is not sliding — it is not moving in a straight line. It is not flipping — it is not producing a mirror image. It is turning about a fixed point. In mathematics, we call this a ROTATION. Write the word ROTATION at the top of your Model Journal page and underline it.' Close the lesson: 'Next lesson we will ask: how much has the	CUMULATIVE MODEL JOURNAL (Science and Engineering Practice: Developing and Using Models). The Model Journal is a running artefact that learners revise across every lesson in the unit. Starting with a simple annotated sketch in Lesson 1 and building towards a full coordinate-geometry model by the final lesson makes the growth of understanding visible and personally owned. The Model Statement writing task requires learners to use precise language to distinguish rotation from translation and reflection — activating prior knowledge of transformations while establishing the conceptual uniqueness of rotation. Returning to the Nairobi clock in the sketch keeps the phenomenon alive as the anchor of the model.	Observable indicator: Every learner's Model Journal page shows a clock face with four hand positions, a labelled centre, a direction arrow, and a written Model Statement. Quality check: the Model Statement must (a) name the motion as 'turning' or 'rotating', (b) mention a 'fixed point' or 'centre', and (c) contrast it with sliding or flipping OR state that size/shape is preserved. Teacher collects 5–6 journals at random to read overnight. Learners who have written 'the hand moves in a circle' without mentioning a fixed centre will receive a targeted prompt at the start of Lesson 2: 'Your model is almost there — what is the one point that never moves?'

		hand turned — and how do we measure that exactly? Look again at the clock images tonight and think: how could you measure the amount of turning?' Record this as a 'To Investigate' prompt at the bottom of the DQB.	
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D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Did the predict phase reveal any strong misconceptions — for example, learners describing the clock hand's motion as a 'slide' or a 'flip'? If so, plan to address these explicitly at the start of Lesson 2 before introducing angle measurement. (2) Review the sticky notes on the DQB. Which pillars attracted the most questions? Use this to prioritise depth in Lessons 2–4. (3) Read the 5–6 collected Model Journal pages. Do learners' Model Statements show a clear distinction between rotation and other transformations? If more than one-third lack the concept of a 'fixed centre', open Lesson 2 with a 60-second repeat of the improvised clock activity, this time explicitly asking: 'What does the pin — the thing that never moves — do to the motion?' (4) Was the improvised clock activity well-resourced? If card arrows were unavailable, note alternatives for next time: a sharpened pencil tied to a string pinned at one end, or a folded strip of newspaper. (5) Did all groups contribute to the DQB? If any group was passive, assign that group the role of 'DQB Curator' in Lesson 2 — responsible for reading all new sticky notes to the class.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed the minute hand of the University of Nairobi Administration Block clock at four positions — 12:00, 12:15, 12:30, and 12:45 — and noted that the hand's length and shape remained identical at every position while its direction changed completely.
What did I learn?	Learners discovered, through the improvised card-arrow clock investigation, that every point on a rotating object stays the same distance from the centre of rotation — and that this type of motion (rotation) is distinct from a translation (slide) or a reflection (flip). Key vocabulary introduced: centre of rotation, angle of rotation, clockwise, anticlockwise.
How does this explain the phenomenon?	Learners explained that the clock hand is performing a rotation about a fixed centre point, that no stretching or flipping occurs, and that the distance from the centre to any point on the hand is preserved. They recorded this as a boxed property in their Rotation Model Journals and posted their initial questions and first partial answers on the class Driving Question Board.

LESSON 2 (80 minutes): Discovering the Properties of Rotation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to physically explore and articulate the four key properties of rotation — congruence of object and image, invariance of the centre, equal angular movement of every point, and the significance of direction — by rotating paper cutouts of plane figures about a fixed pin.
Knowledge	Learners will know that: (a) the object and its image under rotation are congruent; (b) every point on the object moves through the same angle about the centre; (c) the centre of rotation is the only invariant (fixed) point; (d) the direction of rotation (clockwise or anticlockwise) affects where the image ends up.
Skills	Learners will be able to: (a) use a pin, paper cutout, and plain paper to rotate a plane figure through a specified angle about a given centre; (b) measure and verify that corresponding sides and angles of object and image are equal; (c) measure and confirm that each vertex travels through the same angle about the centre; (d) record and communicate observations systematically in a group observation table; (e) discuss and compare the properties of rotation with those of translation and reflection.
Attitudes	Learners will: (a) appreciate the value of careful, systematic observation in building mathematical understanding; (b) respect and value each group member's contribution during collaborative investigation; (c) take responsibility for the safe handling and return of shared learning materials (pins, rulers, protractors, cutouts).
Key Inquiry Question	How is rotation applied in real-life situations?
Purpose in Storyline	In Lesson 1 learners predicted what kind of motion the minute hand of the University of Nairobi clock performs and recorded initial questions on the Driving Question Board. Lesson 2 is the first full OBSERVE & EXPLORE lesson: learners now gather hands-on physical evidence by rotating paper cutouts about a pin (the fixed centre), directly mirroring what the clock hand does. Every property they discover — congruence, fixed centre, equal angle, direction — becomes a verified answer to a question on the DQB, and each property is framed as a rule that governs the clock hand's journey. A connection to the posho mill phenomenon (the rotating millstone turns grain about a fixed axle, preserving the mill's shape while sweeping every point through the same angle) is made explicit at the close of the lesson.
Safety Notes	Drawing pins are sharp. Learners must handle pins with care — always press the pin into the paper/board using a fingertip placed on the flat head, never the point. Pins must be collected and counted back into the container at the end of the activity. No pin should be left loose on desks or the floor. Learners with known sensitivity to sharp objects should use a blunt-ended compass instead of a pin under teacher supervision.

B. LESSON OVERVIEW

Learners work in groups of four or five. Each group receives a paper cutout of a triangle and a quadrilateral, a plain sheet of A3 paper, a drawing pin, a ruler, a protractor, and a pencil. They place the cutout on the plain paper, pin it to a fixed centre point, and rotate it through specified angles (90° , 180° , 270° clockwise and anticlockwise), tracing both object and image each time. Using rulers and protractors they measure corresponding sides, angles, and the angular displacement of each vertex from the centre. Groups record findings in a structured observation table. A whole-class gallery walk and discussion follows, culminating in a co-constructed class list of the Four Properties of Rotation, which is posted on the Driving Question Board. The lesson closes by connecting findings to the University of Nairobi clock (the anchoring phenomenon) and to the posho mill supporting phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Determining rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners are shown the large printed image sequence from Lesson 1 — the University of Nairobi clock tower minute hand at 12:00, 12:15, 12:30, and 12:45 — displayed on the board. Working individually in their exercise books (2 minutes, silent), each learner writes answers to two rapid recall questions: (1) 'In Lesson 1, we said the minute hand performs a special kind of motion — write TWO things you said STAY THE SAME as the hand moves.' (2) 'Write ONE question from our Driving Question Board that you think today's activity might help answer.' Learners then share with a shoulder partner (1 minute), and three pairs are cold-called to share with the class. Learners record their pre-lesson prediction: 'I predict that when I rotate a paper cutout about a pin, the object and its copy will be _____ and the pin point will _____.'	Display the clock image sequence on the board. Say: 'Cast your mind back to Lesson 1. Look at the clock. You are going to write — silently, in your exercise books — two things that STAY THE SAME as the minute hand moves, and one question from our DQB that today might answer. You have two minutes. Begin.' [Allow 2 minutes of silent individual writing. Circulate, noting what learners write — do not prompt yet.] After 2 minutes say: 'Share your two points with your shoulder partner. Thirty seconds.' [Allow 30 seconds.] Then cold-call exactly three pairs: 'Pair at the front-left window — what did you write stays the same?' [Wait 10 seconds before prompting.] Repeat for two more pairs. Affirm correct observations (shape, size, distance from centre) without yet confirming or denying predictions about today's findings. Say: 'Write your prediction for today in the frame on the board: I predict that when I rotate a paper cutout about a pin, the object and its copy will be _____ and the pin point will _____.' [Give 1 minute.] Cold-call one more learner to read their prediction aloud. Record two or three predictions on a sticky note on the DQB under 'Our Predictions for Lesson 2'.	Retrieval Practice + Prediction Frame. Rapid individual retrieval of prior lesson knowledge activates relevant schema before new evidence is gathered. The structured prediction frame ('the object and image will be ____ and the pin will ____') gives learners a concrete claim to test during the activity, ensuring they observe purposefully rather than passively. Connecting predictions to the DQB signals that today's evidence will directly update their growing understanding of the phenomenon.	Teacher scans exercise books during the 2-minute writing period. Observable indicator of readiness: learners can name at least one preserved property (shape, size, or distance from centre) from Lesson 1. If fewer than half the class can do this, teacher pauses and briefly re-shows the Lesson 1 phenomenon clip (30 seconds) before proceeding. Cold-call responses reveal whether learners distinguish between what changes (direction/orientation) and what stays the same (size, shape), which is the conceptual core of today's lesson.
Observe Phase  VIDEO: Proving the SAS triangle congruence	Groups of four or five receive a materials pack: one A3 plain paper sheet, one triangle cutout (labelled ABC) and one quadrilateral cutout (labelled PQRS) made from card, one drawing pin, one ruler (30 cm),	Distribute materials packs (pre-assembled) to each group before the phase begins. Say: 'Your task card has four rounds and four analysis questions. In Round 1 you rotate 90° clockwise. In Round 2	Evidence-Recording Investigation with Structured Analysis Questions. By physically rotating a cutout (rather than reading about rotation), learners generate their own primary data —	Teacher checks group observation tables during circulation. Three specific observable indicators: (1) $OA = OA'$, $OB = OB'$, $OC = OC'$ recorded correctly (distance preservation); (2) angle $AOA' =$

[criterion using transformations](#)

Source: Khan Academy (English - US curriculum)
[Search ARES](#) for similar videos

 EXERCISE:
[Determine rotations \(basic\)](#)

Source: Khan Academy (English - US curriculum)
[Search ARES](#) for similar readings

one protractor, and two pencils. Step-by-step task card guides the investigation across four rounds. Round 1: Place triangle ABC on the A3 paper. Mark a centre point O near the centre of the paper. Press the pin through the centroid of the paper (NOT through the cutout — the pin goes into the paper board beneath, and the cutout is placed over it, with the pin marking the centre O). Trace the triangle in its starting position — label it 'Object'. Rotate the cutout 90° clockwise about O. Trace the new position — label it 'Image 1'. Measure: (a) all three sides of Object and Image 1; (b) all three angles of Object and Image 1; (c) the distance OA and OA', OB and OB', OC and OC'; (d) angle AOA', angle BOB', angle COC' using the protractor. Record all measurements in the group observation table. Round 2: Repeat for 180° clockwise rotation — label 'Image 2'. Round 3: Repeat for 90° anticlockwise rotation — label 'Image 3'. Round 4: Repeat the full investigation using quadrilateral PQRS. After all rounds, each group discusses and answers four analysis questions on the task card: (Q1) 'Are the object and each image congruent? How do you know?' (Q2) 'Did the pin point O move at any point? What does this tell you?' (Q3) 'Compare angle AOA', BOB', and COC' for each rotation. What do you notice?' (Q4) 'Compare Image 1 (90° clockwise) with Image 3 (90° anticlockwise). What is different? What does this tell you about direction?' Groups record answers in full sentences in their exercise books.

you rotate 180° clockwise. In Round 3 you rotate 90° anticlockwise. In Round 4 you use the quadrilateral. You have 28 minutes for all four rounds plus your analysis questions. I am looking for: accurate measurements written in the table, full-sentence answers to the four questions. Begin Round 1 now.' [Circulate every 4–5 minutes. At each group, crouch to their level and ask: 'Show me your measurement for OA and OA' — are they equal? What does that tell you?' Wait 10 seconds for a response before prompting.] At the 10-minute mark, give a time check: 'You should be starting Round 2 now — check your progress.' At the 20-minute mark: 'Rounds 3 and 4 — move on.' At the 28-minute mark: 'Pens down on the activity — complete your analysis questions now. You have 5 more minutes for Q1 to Q4.' [After 5 minutes, say:] 'Stop writing. Each group nominate your spokesperson. We will now do a gallery share.' Ensure every group has answered Q3 — this is the most mathematically significant observation (equal angular displacement). If any group has not reached it, prompt: 'Look at angle AOA' and angle BOB' in your table — are they the same number or different numbers?'

measurements of sides, angles, and angular displacements. The four structured analysis questions (Q1–Q4) scaffold the cognitive move from raw data ('OA = 5 cm, OA' = 5 cm') to mathematical generalisation ('every point stays the same distance from the centre'). Embedding direction comparison (Q4) within the same activity prevents a common misconception that clockwise and anticlockwise rotations through the same angle always produce the same image. The quadrilateral in Round 4 broadens the evidence base beyond triangles, beginning to establish that the properties are general, not shape-specific.

angle BOB' = angle COC' recorded as equal values (equal angular displacement); (3) Image 1 and Image 3 traced in visibly different positions on the paper (direction sensitivity). Any group that has Image 1 and Image 3 overlapping has not changed direction — teacher intervenes: 'For Image 3 you should have rotated anticlockwise — which way is anticlockwise if you are looking at a clock face?' [Wait 12 seconds.] Groups unable to articulate Q1 in writing are asked: 'Use your ruler now — measure AB and A'B'. Are they the same length? What is the word for figures that have all sides and angles equal?'

Explain Phase  VIDEO: Proving the ASA and AAS triangle congruence criteria using transformations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	A gallery share: each group's A3 rotation sheet is held up or taped to the board/wall by the spokesperson. The class observes all groups' sheets for 2 minutes (silent gallery walk around the room or from their seats if space is limited). Learners use a sticky dot or pencil tick to mark any other group's sheet that shows a result that surprises them. Whole-class discussion follows. The teacher facilitates a structured discussion guided by the four analysis questions, cold-calling from different groups for each. As each property is confirmed by multiple groups' evidence, the teacher writes it on a large chart paper in agreed class language — this becomes the Class List of Properties of Rotation. Learners copy the list into their exercise books under a heading 'What we discovered: Properties of Rotation.' The teacher then explicitly connects each property back to the University of Nairobi clock: 'Property 1 says the object and image are congruent — what does that mean for the minute hand? Is the hand at 12:15 a different size or shape from the hand at 12:00?' Learners respond. Then the supporting phenomenon connection: 'Think about the millstone at a posho mill in Kericho — it rotates about the axle. Which of our four properties applies to the millstone?' Learners discuss in pairs (1 minute) then three pairs are cold-called.	After the activity phase, say: 'Hold up your A3 sheet — I want everyone to see all groups' work. If you are in a fixed seat, look at the sheets being held up. You have 90 seconds to look and put a mental tick on anything surprising.' After 90 seconds: 'Spokesperson for Group 1 — read me your answer to Q1.' [Wait 10 seconds.] After response: 'Group 3, do you agree or disagree — and what is your measurement evidence?' [Wait 10 seconds.] Write on chart paper: 'Property 1: The object and its image under rotation are CONGRUENT — corresponding sides and angles are equal.' Underline 'CONGRUENT.' Repeat the cold-call pattern for Q2 (invariant centre), Q3 (equal angular displacement), and Q4 (direction matters). For Q3, specifically ask: 'Group 4, what was angle AOA' for your 90° rotation?' [Wait 10 seconds.] 'And angle BOB'?' [Wait 10 seconds.] 'And angle COC'?' [10 seconds.] 'What pattern do you see?' After all four properties are on the chart, say: 'Now — the University of Nairobi clock. The minute hand moves from 12:00 to 12:15. Property 1: is the hand at 12:15 the same shape and size as at 12:00? Call out — yes or no.' [Class responds.] 'Property 2: did the centre of the clock move? Call out.' [Class responds.] 'Property 3: did every point on the hand move through the same angle — 90°? Call out.' 'Property 4: which direction does the clock hand move — clockwise or anticlockwise? Why does direction matter?' Cold-call two learners for Property 4. Then: 'Turn to your partner. Tell them: which of our	Gallery Share + Whole-Class Evidence Synthesis. By exposing learners to multiple groups' physical evidence (six different A3 sheets) before discussing, the class builds confidence that the properties are not artefacts of one group's measurements but consistent findings across different figures and different rotation angles. The teacher's explicit language moves ('Property 1... Property 2...') formalise learner-generated observations into canonical mathematical vocabulary without replacing learner agency. Connecting each property to the clock phenomenon and the posho mill creates two anchors — one familiar urban (University of Nairobi) and one rural-productive (Kericho posho mill) — making the abstraction culturally and contextually accessible.	During the structured discussion, teacher listens for three quality indicators: (1) Learners use the word 'congruent' (not just 'same') by the end of Property 1 discussion — if not, teacher re-prompts: 'We have a mathematical word for figures with equal sides and equal angles — what is it?' [10 seconds wait.] (2) At least one learner spontaneously links equal angular displacement (Property 3) back to the clock ('so every mark on the minute hand moves 90° when it goes from 12 to 3') — teacher notes this and highlights it. (3) During the posho mill pair discussion, teacher circulates and checks that learners are identifying a minimum of two properties (not just one) for the millstone. Any pair identifying only one property is prompted: 'What about Property 2 — where is the axle of the millstone? Does it move?'
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		four properties applies to the millstone at a posho mill? One minute.' Cold-call three pairs.		
Driving Question Board (DQB) Creation  VIDEO: Proving triangle congruence Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner receives two sticky notes (or writes on a slip of paper). On the first sticky note (green), the learner writes one thing from today's investigation that directly answers a question already on the DQB. On the second sticky note (orange), the learner writes one new question that today's lesson has made them wonder about (e.g., 'How do we calculate the exact position of A' if we know the angle and the coordinates of A?' or 'What happens if the centre of rotation is inside the shape?'). Learners place their green notes on the DQB next to the question they believe they have answered, and orange notes in a 'New Wonders' section. The class reads two or three of each aloud. The teacher also adds the co-constructed Class List of Properties (a summary card) to the DQB as 'Evidence from Lesson 2'. Groups briefly discuss: 'Looking at our DQB — what big question remains unanswered that we still need to figure out?'	Distribute two sticky notes to each learner. Say: 'On the GREEN note, write one thing you proved today that answers a question already on our DQB. On the ORANGE note, write one new question that today has made you wonder about. You have 3 minutes — write now.' [Circulate. After 3 minutes:] 'Come and place your green note next to the DQB question it answers. Place your orange note in the New Wonders section.' [Allow 2 minutes for placement.] Read two green notes aloud — choose ones that use the word 'congruent' or 'invariant' to model precise mathematical language. Read two orange notes aloud — choose ones that point forward to Lesson 3 content (coordinates, calculation of image position). Say: 'I am adding our Class List of Properties to the DQB — this is our shared evidence from Lesson 2.' Pin the chart paper summary card. Ask: 'Look at the whole DQB. What ONE big question is still completely unanswered?' Cold-call three learners. [Wait 10 seconds each.] Affirm responses pointing to the calculation of image coordinates — 'That question — how do we find exactly where A' is using numbers — is what Lessons 3 and 4 will tackle.'	Two-Colour Sticky Note DQB Protocol. Using two distinct colours (evidence vs. new wonders) trains learners to distinguish between claims they can now support with evidence and questions that require further investigation. This metacognitive routine makes the incremental, evidence-based nature of scientific and mathematical thinking visible and tangible. Publicly placing evidence notes next to the specific DQB question they answer gives learners the experience of 'closing a loop' — a powerful motivator for continued inquiry. The teacher's selection of forward-pointing orange notes creates a productive bridge to Lesson 3.	Teacher reads all green sticky notes after the lesson (not in real time) to check three things: (1) Does the learner correctly identify which DQB question their evidence addresses? (2) Does the learner's language reflect today's formal properties (congruence, invariant centre, equal angle, direction)? (3) Does the learner's orange note suggest they are thinking beyond today's content toward calculation/coordinates (readiness for Lesson 3) or are they still clarifying today's concepts (needs consolidation before Lesson 3)? Any learner whose green note contains a misconception (e.g., 'rotation makes the shape bigger') is noted for targeted follow-up at the start of Lesson 3.
Model Building Phase  VIDEO: Proving the SSS triangle congruence	In Lesson 1, learners drew an initial annotated sketch model of the University of Nairobi clock, labelling what they thought stays the same and what changes. They now retrieve that sketch from their exercise books. Using a different	Say: 'Open your exercise books to your Lesson 1 clock sketch model. Pick up a different colour pen — red, blue, or green — something different from what you used in Lesson 1. You are going to UPGRADE your model using	Colour-Coded Cumulative Model Revision. Using a visually distinct colour to revise the Lesson 1 model makes the growth of understanding literally visible on the page — learners can see exactly what new knowledge today	Teacher checks two specific features in the revised models before learners close their books: (1) Arcs must be drawn from at least two different points on the hand (not only the tip) — this verifies understanding of Property

criterion using transformations Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum) Search ARES for similar readings	coloured pen (to show revision), they update the model in light of today's evidence by: (a) adding the label 'CENTRE OF ROTATION = fixed / invariant' to the pivot point of the clock; (b) drawing arcs from two or three specific points on the minute hand to their new positions (at 12:15, 12:30, 12:45) and labelling each arc '90°'; (c) writing the four properties in a box beside the clock sketch, using today's class-agreed language; (d) adding a new element: a small sketch of a posho mill millstone with arrows showing its rotation and the axle labelled as 'centre of rotation — invariant'. Learners write one sentence below the model: 'What the clock hand and the millstone have in common is _____. ' Two learners share their completed revised models with the class.	today's evidence. You have 7 minutes. Add four things: (1) label the centre as invariant; (2) draw arcs showing 90° of movement for two points on the hand; (3) add a box with your four properties; (4) sketch the posho mill millstone and label the axle as centre of rotation. Write the sentence at the bottom: 'What the clock hand and the millstone have in common is _____. Begin.' [Circulate for 7 minutes, checking that arcs are drawn from specific points — not just from the tip of the hand — and that '90°' is labelled on each arc. If a learner has only drawn arcs from the tip, prompt: 'What about a point in the middle of the hand — does it also move 90°?' Wait 12 seconds.] After 7 minutes: 'Two volunteers to show their model and read their sentence.' [Choose one stronger and one mid-range model.] After both share, summarise: 'Every time we add to our model, we are getting closer to answering our driving question: what mathematical rules govern exactly where every point on the hand ends up. Today we know four properties. In Lesson 3 we will start calculating exact positions using coordinates.'	added to what they already had. Drawing arcs from multiple points on the hand (not just the tip) directly operationalises Property 3 (equal angular displacement) in a concrete visual form. Adding the posho mill to the same model page reinforces the cross-context generality of the four properties — rotation is not only what clock hands do; it is a universal geometric transformation that applies wherever a rigid object turns about a fixed point. The sentence-completion frame ('What the clock hand and the millstone have in common is _____.') prompts learners to articulate the abstraction in their own words, deepening conceptual consolidation.	3 (equal angular displacement for ALL points, not just the tip). (2) The four properties box must include the word 'congruent' and the phrase 'invariant' or 'does not move' for the centre. Learners missing either element are asked to add it before closing their books. The sentence completion ('What the clock hand and the millstone have in common') is collected on an exit slip (learners tear or cut the sentence out and hand it to the teacher as they leave) for rapid whole-class diagnostic review before Lesson 3.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. EVIDENCE OF UNDERSTANDING: When learners measured angle AOA' and angle BOB' during Round 1, did they find equal values? If several groups found unequal values, what was the likely source of error — measuring technique, protractor placement, or conceptual misunderstanding? How will you address this at the start of Lesson 3?
2. LANGUAGE PRECISION: By the end of the Explain phase, were learners using the word 'congruent' spontaneously, or did you have to supply it every time? If you had to supply it repeatedly, plan a 2-minute vocabulary warm-up for Lesson 3 using the word in a Kenyan context (e.g., 'The two chapati rounds cut from the same

dough — are they congruent?').

3. DIRECTION SENSITIVITY: Did learners clearly distinguish between clockwise and anticlockwise rotations producing different images (Property 4)? A common finding is that weaker groups place Image 1 and Image 3 in the same position. Note how many groups made this error and whether targeted intervention in Lesson 3 is needed before introducing coordinate-based rotation.

4. POSHO MILL CONNECTION: When learners discussed the Kericho posho mill supporting phenomenon, how many properties did they spontaneously identify (target: at least two per pair)? If learners struggled, consider bringing in a photograph of a posho mill millstone for the Lesson 3 wall display alongside the clock image.

5. DQB QUALITY: Review the orange 'New Wonders' sticky notes collected. Are the new questions pointing forward to calculation of image coordinates (ideal — ready for Lesson 3)? Or are they still asking about the basic meaning of rotation (needs consolidation)? Adjust the Lesson 3 opening accordingly.

6. INCLUSION & PACING: Were all group members — including quieter learners and any learners with lower prior attainment — able to participate meaningfully in the measurement activity? Consider assigning explicit roles in Lesson 3 groups (Measurer, Recorder, Checker, Spokesperson) to ensure equitable participation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners physically rotated paper cutouts of triangles and quadrilaterals about a fixed pin through angles of 90° , 180° , and 270° in both clockwise and anticlockwise directions, tracing object and image positions on A3 plain paper and recording measurements of corresponding sides, angles, distances from centre (OA, OA', OB, OB'), and angular displacements (angle AOA', BOB', COC') in a structured group observation table.
What did I learn?	Learners discovered and co-constructed the four properties of rotation: (1) the object and image are congruent — all corresponding sides and angles are equal; (2) the centre of rotation is invariant — it is the only point that does not move; (3) every point on the object moves through exactly the same angle about the centre; (4) the direction of rotation (clockwise vs. anticlockwise) determines where the image ends up, so direction must always be specified. These properties were connected to the movement of the University of Nairobi clock minute hand (the anchoring phenomenon) and to the rotating millstone of a posho mill in Kericho (the supporting phenomenon).
How does this explain the phenomenon?	Learners updated their cumulative annotated sketch model of the University of Nairobi clock, adding colour-coded arcs showing equal 90° angular displacement from multiple points on the minute hand, labelling the clock centre as 'invariant', recording the four properties in a box, and adding a posho mill millstone sketch with its axle labelled as 'centre of rotation'. Learners also updated the Driving Question Board with evidence notes (green) linking today's findings to prior DQB questions and new wonder notes (orange) pointing toward how exact image coordinates can be calculated — the focus of Lesson 3.

LESSON 3 (40 minutes): Rotating Objects on a Plane Surface — Constructing Images with a Geometrical Set

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners develop and practise the step-by-step geometric construction method for rotating any object on a plane surface using a ruler, compass, and protractor, connecting each construction step back to the fixed centre of the University of Nairobi clock tower.
Knowledge	– State and apply the four-step construction method for rotating an object on a plane surface given a centre and angle of rotation. – Distinguish between rotations with the centre inside, outside, and on the boundary of the object, and predict how each affects the image position. – Connect the constructed image to the four properties of rotation established in Lesson 2 (congruence, invariant centre, equal angle, specified direction).
Skills	– Accurately construct the image of a given polygon under a specified rotation (centre and angle: 90°, 180°, 270°) on plain paper using a ruler, compass, and protractor. – Measure and verify that each image vertex lies at the correct distance from the centre and at the correct angle from the corresponding object vertex.
Attitudes	– Appreciate the precision and systematic thinking required in geometric construction, recognising that a small measurement error changes the entire image. – Show respect for peers' construction work by offering specific, evidence-based feedback during pair-checking.
Key Inquiry Question	How is rotation applied in real-life situations?
Purpose in Storyline	Having discovered what rotation is (Lesson 1) and its four properties (Lesson 2), learners now learn HOW to produce a rotated image precisely — building the construction toolkit they will later apply on the Cartesian plane. Every construction step is explicitly linked back to the clock at the University of Nairobi: the compass pin represents the fixed clock centre; the protractor angle represents the sweep of the minute hand.
Safety Notes	Compass points are sharp. Remind learners to place the compass on the desk when not in use and never to point it at a peer. Protractors and rulers should be handled carefully to avoid breakage.

B. LESSON OVERVIEW

This lesson is the first full construction lesson in the rotation sequence. Building directly on the four properties co-constructed in Lesson 2, learners move from understanding what rotation does to knowing precisely how to produce it on paper. Each learner uses a geometrical set — ruler, compass, and protractor — to construct the image of a given triangle or quadrilateral under a specified rotation, working through a collaboratively developed four-step method: (1) mark the centre of rotation; (2) draw a ray from the centre through each vertex; (3) use the compass to mark the image vertex at the same distance from the centre along a new ray; (4) measure the given angle at the centre using the protractor to position that new ray correctly.

Three distinct centre positions are explored through a sequence of worked examples: the centre inside the object (analogous to a wheel spinning on its own axle — like the wheel of a matatu on Ngong Road), the centre outside the object (analogous to a ball on a string swung in a circle — like a sling used by herdsman in the Rift Valley), and the centre on the boundary of the object. This variation is crucial because it challenges the misconception that the centre must be in the middle of the shape. For each position, learners predict where the image will land before constructing, then verify by construction.

The anchoring phenomenon — the University of Nairobi clock — is the explicit scaffold throughout: the teacher returns to it at every transition, asking learners to identify which part of the construction corresponds to which feature of the clock (the compass pin = the clock's fixed centre; the compass arc = the circular path traced by any point on the minute hand; the protractor angle = the angular sweep of the hand). By the end of the lesson, learners can construct a rotation image confidently and articulate why each step is mathematically necessary, advancing the driving question from "what are the rules?" to "how do we apply the rules to find any new position?"

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Determining rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner receives a plain sheet of paper with a pre-drawn triangle ABC (labelled with vertices) and a clearly marked point O outside the triangle, labelled 'Centre of Rotation.' Learners are asked — without any tools yet — to sketch freehand where they think triangle ABC will end up after a 90° anticlockwise rotation about point O. They annotate their sketch with a sentence explaining their reasoning, then share their prediction with a shoulder partner.	Distribute the prediction sheets face-down. Say: 'Before you turn the sheet over, I want you to picture the clock at the University of Nairobi's Administration Block. Imagine the minute hand sweeping 90° — from 12 to 9. Now turn your sheet over.' Allow 10 seconds of silent looking. Then say: 'Without picking up a ruler or compass yet, sketch where you think this triangle will end up after a 90° anticlockwise rotation about point O. Write one sentence: why does your image land there?' Allow 3 minutes of individual silent work. Circulate and note 2–3 contrasting predictions — one that keeps the shape too close to O, one that moves it far, one that approximately correct. After 3 minutes, cold-call 3 learners: 'Amina, show us your sketch and read your sentence.' 'Brian, yours looks different — what was your reasoning?' 'Chepkirui, what is the same in all three predictions?' Wait 12 seconds after each question before accepting a response. Do NOT correct predictions yet — affirm all responses with 'Interesting — let's keep that in mind as we build our method today.'	Prediction with Justification — learners externalise their current mental model of rotation before instruction, creating a cognitive anchor. Comparing contrasting predictions activates productive confusion (why do our answers differ?) that the Observe phase will resolve through construction.	Teacher scans sketches for two key misconceptions: (1) the learner has translated (slid) the triangle rather than rotating it — image is same orientation, just shifted; (2) the learner has reflected the triangle. Note names of learners showing these misconceptions for targeted questioning during the Explain phase. A correct prediction shows the triangle in a new orientation with each vertex equidistant from O as the original.
Observe	The teacher demonstrates the	Say: 'We are going to build our	Gradual Release with Live	During Step 3, check that learners

Phase  VIDEO: Disc method rotation around horizontal line Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations (basic) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	four-step construction method live at the board using a large printed triangle and a giant protractor and compass (or on a whiteboard with coloured markers). Learners follow along on their own sheets, constructing each step simultaneously. The four steps are displayed as a visible anchor chart titled 'The Clock Construction Method.' After the demonstration, learners independently attempt a second example — a quadrilateral rotated 180° clockwise about a centre inside the shape — and observe what happens to the image orientation.	construction method step by step — and at every step, I will ask you: which part of the Nairobi clock does this represent?' STEP 1 — Mark the centre: 'Place your compass pin on point O. This pin is the clock's fixed centre — the axle. It will not move.' Draw O clearly on the board. STEP 2 — Draw a ray from O through each vertex: 'Use your ruler to draw a straight ray from O through vertex A, extending it beyond A. Now through B. Now through C. These rays are like the clock's radius lines — they radiate outward from the fixed centre.' Allow 90 seconds for learners to complete. Circulate and check 5 learners' rays are extended beyond the vertex, not stopped at it. STEP 3 — Measure the distance: 'Open your compass to the distance OA. This distance is the length of our clock hand from axle to tip — it stays the same after rotation. Mark a small arc on a new ray. You will come back to this arc in Step 4.' Demonstrate. Allow 2 minutes. Cold-call: 'Kipchoge — why must OA equal OA-prime?' Wait 10 seconds. Expected answer: 'Because rotation preserves distance from the centre.' STEP 4 — Measure the angle: 'Place your protractor at O with its baseline along ray OA. Measure 90° anticlockwise. Draw a new ray at that angle. Where this new ray crosses your compass arc — that	Modelling — the teacher models each step explicitly, naming its connection to the phenomenon (clock), then immediately releases learners to execute that step. This 'I do, we do, you do' structure within a single demonstration sequence ensures procedural encoding is linked to conceptual meaning at every micro-step, not just at the end.	re-open the compass for each vertex individually (OA, OB, OC may differ) — a common error is keeping the compass at the same opening for all vertices. During Step 4, check that the protractor baseline is aligned to the ray from O through the vertex, not to a horizontal. During independent practice, look for: (a) image vertices correctly positioned at equal distances from O; (b) the 180° image is a 'half-turn' — the shape appears upside-down/rotated, not reflected.
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		is A-prime, the image of A.' Demonstrate slowly. 'Repeat for B and C, then connect A-prime, B-prime, C-prime.' Allow 4 minutes for learners to complete the full construction. Anchor chart is left on the board for the rest of the lesson. Say: 'Now try this alone: quadrilateral PQRS, centre O inside the shape, rotate 180° clockwise. Use the same four steps.' Allow 5 minutes. Circulate actively — do not correct, but ask: 'Which step are you on? Show me how you measured your angle.'		
Explain Phase  VIDEO: Determining rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners work in pairs to complete three targeted practice constructions on plain paper, each with a different centre position: Example A — equilateral triangle, centre outside (like the satellite dish rotating on the rooftop of a building on University Way, Nairobi); Example B — rectangle, centre on one side of the shape (like a door rotating on its hinges at the school entrance); Example C — irregular quadrilateral, centre inside the shape (like the rotating blades of a maize-grinding posho mill in Kericho). For each, they rotate through 90°, 180°, or 270° as specified on their worksheet. After constructing, pairs write a one-sentence comparison: 'When the centre is outside the shape, the image is _____ compared to when the centre is inside because _____.'	Distribute the three-example worksheet. Say: 'Each example has a different centre position. I want you to notice — does the centre position change how you do the construction, or only where the image ends up?' Pair learners (stronger constructors paired with learners who showed translation/reflection errors in the Predict phase). After 8 minutes of pair work, stop the class. Cold-call 3 pairs — one per centre position: 'Zawadi and Mwenda — you had the centre outside. Hold up your construction. Class, does the image size match the object? How do you know?' Wait 12 seconds. 'Aisha and Silas — centre on the boundary. What happened to the vertex that was right on the centre?' (Expected: it did not move — it is invariant.) Wait 10 seconds. 'Baraka and Njeri — centre inside. Did the image overlap with the object?'	Variation Theory — the three examples are deliberately designed to vary only the centre position while keeping everything else (same construction steps, same angle used) constant. This isolates the effect of centre position on the resulting image, helping learners disentangle the procedure from the outcome and deepening conceptual understanding of why the centre matters.	Collect the three constructions at the end of the pair activity. Check: (1) for Example B (centre on boundary), the vertex sitting on the centre must be unmoved in the image — this is a direct application of the invariant centre property; (2) for Example C (270° rotation), the image should appear in the correct quadrant — a 270° anticlockwise rotation is equivalent to a 90° clockwise rotation, and teacher notes which learners have caught this relationship. The one-sentence comparison reveals whether learners understand that the procedure is universal but the image position varies with centre location.

		Now address the prediction sheets from Phase 1: 'Look at your sketch from the beginning of the lesson. Put a tick next to anything your prediction got right. Put a question mark next to anything the construction surprised you about.' Allow 2 minutes. Cold-call 2 learners: 'What surprised you?' Connect explicitly to the phenomenon: 'Every single step of our construction — the compass arc, the protractor angle, the fixed pin — is exactly what the clock at the University of Nairobi does every minute. The compass pin is the clock's centre. The arc is the path of any point on the hand. The angle is the minute hand's sweep. We have just described the clock's mathematics precisely enough to reproduce it on paper.'		
Driving Question Board (DQB) Creation  VIDEO: Determining rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners individually write one updated claim and one new question on sticky notes (or index cards) in response to the prompt: 'We can now say we KNOW: _____. But we are still wondering: _____.' Cards are added to the class Driving Question Board under two columns: 'What we can now do' and 'What we still need to figure out.' The teacher reads several claims and questions aloud and facilitates a 3-minute whole-class discussion to prioritise the most important unanswered questions for future lessons.	Display the driving question prominently: 'When the minute hand of the clock at the University of Nairobi completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up — and how can we use those rules to rotate any shape to any position, on paper and on a grid?' Say: 'Today we learned HOW to construct a rotation on plain paper. Look at the driving question again — what part of it can we now answer? And what part are we still missing?' Allow 2 minutes of silent individual writing. Collect cards and read 4–5 aloud. Post the following anticipated new questions on the board (generated by the teacher if not raised by learners): 'How do we rotate on a	Claim-Question Tracking — by requiring learners to articulate both what they now know and what they still wonder, the DQB update externalises metacognitive progress. It positions knowledge as provisional and growing, motivating the next lesson's investigation and maintaining the storyline's forward momentum.	Teacher reads all submitted 'I KNOW' claims. Red-flag any claim that overgeneralises (e.g., 'the image is always on the opposite side of the centre') or that confuses rotation with reflection. These learners will receive a targeted correction card at the start of Lesson 4. Teacher also notes whether learners are asking questions about the coordinate/formula aspect — evidence that they are ready for Lesson 4's content.

		grid with coordinates instead of plain paper?', 'Can we write a formula instead of always drawing?', 'How do we find the centre and angle if we are only given the object and image?' Say: 'These questions will drive our next lessons. Notice how today's lesson moved our answer forward — we can now produce the image on plain paper. Next, we tackle the grid.' Circle the question about coordinates on the DQB and say: 'That is our target for Lesson 4.'		
Model Building Phase  VIDEO: Determining rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner retrieves their cumulative rotation model — the annotated diagram of the University of Nairobi clock face that has been built across Lessons 1 and 2. They now add a new layer: using a compass and protractor directly on their model diagram, they construct the exact position of the clock's minute hand at 12:15 (90° clockwise from 12:00) and at 12:45 (270° clockwise, or equivalently 90° anticlockwise). They label the constructed image tip as A' and A" respectively, draw the compass arc showing the circular path, and annotate: 'Each construction step I used today is visible here — pin at centre O, arc showing constant distance, protractor angle showing 90°.' They then write a two-sentence 'Model Update Statement' at the bottom of their diagram.	Say: 'Take out your clock model from Lessons 1 and 2. You have drawn the clock and labelled its properties. Now you are going to use today's construction skill directly on that model.' Distribute compasses and protractors. Allow 5 minutes for individual construction on the model. Circulate and ask targeted questions to 4–5 learners: 'Where is your centre of rotation on this clock model?' (Expected: centre of the clock face, point O.) 'What is the radius of your compass set to?' (Expected: the length of the minute hand.) 'You have drawn the hand at 12:00 and 12:15 — what angle did you measure?' (Expected: 90°.) After construction, say: 'Write your Model Update Statement. It must include: what you CAN now do with the clock model that you could not do at the start of Lesson 3, and what question you still need to answer to fully solve the driving	Cumulative Model Revision — returning to the same physical artefact (the clock model) in every lesson creates a visible record of growing understanding. Adding new construction layers to the existing diagram makes the progression from observation (Lesson 1) to properties (Lesson 2) to construction (Lesson 3) literally visible on a single page, reinforcing the coherence of the unit's storyline.	Collect model diagrams at the end of the lesson. Assess three things: (1) the constructed hand positions at 12:15 and 12:45 are geometrically accurate — compass arc visible, protractor angle correctly measured at O; (2) the Model Update Statement uses correct vocabulary (centre, angle, clockwise/anticlockwise, image, distance); (3) the learner's stated 'still need to answer' question points toward coordinates or formulae — evidence of readiness for Lesson 4. Diagrams showing hand positions constructed as straight lines without arcs (missing the compass step) indicate the learner has learned the procedure superficially and will need revision at the start of Lesson 4.

		question.' Cold-call 2 learners to read their statements aloud. Affirm and connect: 'You have just done exactly what a structural engineer does when designing a rotating gate or a Ferris wheel — you have used precise geometric construction to predict where every point ends up. The mathematics you used today is the same mathematics behind every rotating machine in Kenya, from the millstone in Kericho to the turbines at Olkaria Geothermal Plant.'	
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D. TEACHER REFLECTION

1. During the Observe phase, how many learners aligned the protractor baseline to a horizontal line rather than to the ray from O through the vertex — and what quick intervention can I use at the start of Lesson 4 to address this systematic error before it transfers to grid work?

2. Did the three centre-position examples (outside, on boundary, inside) produce the 'noticing' I intended — could I see learners adjusting their mental model of the image location, or did most treat all three as the same construction?

3. When I cold-called learners to explain why OA must equal OA-prime, did their language reference the property from Lesson 2 (invariant distance from centre), or were they reciting a procedural rule without conceptual grounding? What does this tell me about the depth of understanding built in Lesson 2?

4. Did the connection to the University of Nairobi clock feel authentic and meaningful to learners during the Explain phase, or did it feel like a forced analogy? How could I strengthen the phenomenon link in Lesson 4 to make the coordinate-based rotation feel equally purposeful?

5. Looking at the DQB new questions, are learners' 'still wondering' items genuinely pointing toward the driving question (coordinates, formulae, finding the centre from object and image), or are they procedural questions about how to use the protractor? If the latter, what scaffold do I need to provide before Lesson 4?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that using a compass pin at the fixed centre O, a compass arc preserving the distance from O to each vertex, and a protractor measuring the given angle produces a precise image of any polygon — and that changing the position of the centre (inside, outside, or on the boundary of the object) changes where the image lands but does not change the four-step construction method.
What did I learn?	The four-step construction method for rotating an object on a plain surface: (1) mark the centre of rotation; (2) draw rays from the centre through each vertex; (3) use a compass to preserve the distance from centre to each vertex; (4) use a protractor to measure the specified angle at the centre and mark the image vertex. The centre is always the only invariant point, regardless of its position relative to the object.
How does this explain the phenomenon?	The construction method directly models what the University of Nairobi clock's minute hand does: the compass pin represents the fixed clock centre, the compass arc represents the constant length of the hand from axle to tip, and the protractor angle represents the angular sweep of the hand. Constructing the hand at 12:15 (90° clockwise) and 12:45 (270° clockwise) on the clock model showed that the same four-step method predicts the exact position of every point on the hand — beginning to answer the driving question about the mathematical rules governing where every point ends up.

LESSON 4 (40 minutes): Rotation on the Cartesian Plane — Discovering and Applying Coordinate Rules

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners transfer the construction skill from plain paper to the Cartesian plane, discover algebraic coordinate rules for rotations of 90° , 180° , and 270° about the origin and other centres, and apply those rules to calculate new positions of points — connecting back to the Nairobi clock tower phenomenon and the Kasarani football.
Knowledge	– State and apply the coordinate rule for a 90° anticlockwise rotation about the origin: $(x, y) \rightarrow (-y, x)$ – State and apply the coordinate rule for a 90° clockwise rotation about the origin: $(x, y) \rightarrow (y, -x)$ – State and apply the coordinate rule for a 180° rotation about the origin: $(x, y) \rightarrow (-x, -y)$ – State and apply the coordinate rule for a 270° anticlockwise ($= 90^\circ$ clockwise) rotation about the origin: $(x, y) \rightarrow (y, -x)$ – Rotate a point or polygon about a centre other than the origin by translating the centre to the origin, applying the rule, and translating back
Skills	– Plot object and image coordinates accurately on a Cartesian grid and verify congruence of object and image – Identify and record coordinate transformation rules through guided pattern-spotting in a table of input–output coordinate pairs – Apply coordinate rules to calculate image positions of marked points on a rotating object (e.g., a football or clock hand)
Attitudes	– Appreciate that mathematics provides exact, predictable rules that describe real physical motion such as a spinning clock hand or a rolling football – Show responsibility and precision when plotting coordinates, recognising that a single plotting error invalidates the entire image
Key Inquiry Question	How is rotation applied in real-life situations — and how do coordinate rules on the Cartesian plane allow us to calculate exactly where every point ends up after a rotation?
Purpose in Storyline	In Lessons 1–3 learners discovered the properties of rotation and practised constructing images using a geometrical set on plain paper. Lesson 4 now bridges physical construction and algebraic rule: learners plot rotations on a coordinate grid, spot the numerical pattern in how coordinates change, and codify it into a rule. This rule is the mathematical engine that answers the driving question — it is no longer enough to draw the answer; we can now calculate it. The Kasarani football revisit makes the rule immediately useful in a Kenyan sporting context.
Safety Notes	No specific hazards. Ensure rulers and pencils are used carefully. Compasses from the geometrical set should be handled with care and kept closed when not in use.

B. LESSON OVERVIEW

Lesson 4 marks the pivotal transition from geometric construction to algebraic rule in the rotation storyline. Having spent Lesson 3 using a compass and protractor to rotate triangles and quadrilaterals on plain paper, learners now bring those same shapes onto a coordinate grid. The move to the Cartesian plane is deliberately motivated by a concrete problem: if a marked spot on a football rolling along the Kasarani Stadium pitch starts at a known grid position, can we calculate — without drawing anything — exactly where that spot will be after the ball rotates 90° or 180° ? This question gives the lesson its purpose and keeps it tethered to the anchoring phenomenon of the University of Nairobi clock tower, whose minute-hand tip traces a perfect arc whose endpoints can equally be described as coordinate pairs.

The lesson is structured around a guided pattern-spotting investigation. Working in pairs, learners rotate a simple triangle about the origin through 90° anticlockwise, 90° clockwise, and 180° , each time recording the before-and-after coordinates of every vertex in a table. The teacher then leads a whole-class comparison of the tables,

prompting learners to articulate what operation was performed on x and on y in each case. From this inductive process the four key rules emerge as learner-generated discoveries rather than teacher-delivered facts, making them far more memorable. The rules are then tested against a rotation about a non-origin centre — a point representing the axle of the Kasarani football — using the translate-apply-translate-back strategy.

Throughout the lesson the teacher uses deliberate wait time (minimum 15 seconds after each higher-order question), cold-calls at least three different students per phase, and circulates to check coordinate tables before the class discussion. The Driving Question Board is updated to reflect that the class can now not only describe rotation qualitatively but also calculate it algebraically — a major advance in answering the driving question about what mathematical rules govern where every point on the Nairobi clock hand ends up.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Rotating shapes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations (basic) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner receives a small printed or hand-drawn Cartesian grid (axes from -6 to 6) with a right-angled triangle ABC plotted at A(1, 1), B(4, 1), C(1, 3). Learners are asked, individually and in silence, to mark with a cross where they predict vertex A(1, 1) will land after a 90° anticlockwise rotation about the origin. They then write a one-sentence justification for their prediction in their exercise books. After 2 minutes, learners share predictions with a shoulder partner.	Display the grid on the chalkboard with triangle ABC clearly labelled. Say: 'Do not use your geometrical set yet — this is a prediction, not a construction. Think back to what we know about rotation: the distance from the centre stays the same. Use that idea and your gut instinct to mark where A(1, 1) will land after a 90° anticlockwise turn about the origin.' Allow 15 seconds of silence after giving the instruction before learners begin writing. After 2 minutes of individual work, say: 'Turn to your neighbour. You have 1 minute — compare your cross and your justification.' Cold-call 3 students (choose one who predicts correctly, one who predicts a reflection, one who predicts a translation) and ask each: 'Tell us where you placed your cross and why.' After each response, do NOT confirm or deny — say: 'Interesting. Let's hold that prediction and find out.' Record all three predicted coordinates on the board in a 'Prediction Column' of a table headed: Vertex Predicted Image Actual Image.	Activate Prior Knowledge Prediction: By asking learners to predict before using tools, the teacher surfaces existing mental models (and common misconceptions — e.g., confusing rotation with reflection across the y-axis). Recording predictions publicly creates accountability and a later 'reveal' moment that strengthens memory consolidation.	Teacher looks for: (1) Are predictions geometrically reasonable — is the predicted image roughly the same distance from the origin as the object point? (2) Are any learners simply reflecting across an axis (a common error at this stage)? (3) Can learners give a distance-based justification, showing they have internalised the property from Lesson 1? Flag learners who place A at $(-1, 1)$ — they are confusing 90° anticlockwise rotation with reflection in the y-axis — for targeted questioning during the Observe Phase.
Observe Phase  VIDEO:	Learners work in pairs using their geometrical sets. They rotate triangle ABC about the origin	Distribute the pre-drawn table (or instruct learners to draw it). Say: 'Use the exact four-step method	Data Table Pattern-Spotting (Inductive Reasoning): The structured table forces learners to	Teacher looks for: (1) Do image coordinates satisfy the distance-preservation property — i.e., does

Rotating shapes Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum) Search ARES for similar readings	through (a) 90° anticlockwise, (b) 90° clockwise, and (c) 180°, constructing each image using the four-step method from Lesson 3. For each rotation they record the image coordinates of all three vertices — A', B', C' — in a pre-drawn table in their exercise books. The table has columns: Original Point 90° ACW Image 90° CW Image 180° Image. After completing the constructions, each pair is given a second task: circle any numerical pattern they notice between the original coordinates and each image column.	we used in Lesson 3 — rays from the centre, compass arc to preserve distance, protractor to measure the angle. I will give you 8 minutes. Both partners must have the constructions in their own books.' Set a visible timer. Circulate to at least 6 pairs; at each pair check: (i) Are rays drawn through each vertex? (ii) Is the compass arc centred on the origin? (iii) Is the protractor aligned correctly? If a pair has placed their protractor baseline along the x-axis instead of along the ray, say: 'Your protractor baseline must sit along the ray from the origin to the vertex — not along the x-axis. Adjust and redo vertex A first, then check B and C.' After 8 minutes, say: 'Stop constructions. Now look at your completed table. With your partner, circle anything that looks like a pattern — how did the x-coordinate change? How did the y-coordinate change? You have 3 minutes.' Cold-call 3 pairs to share one pattern each. Record responses verbatim on the board under a heading 'Patterns We Noticed.'	place object and image coordinates side by side, making the algebraic relationship visible as a pattern before it is formalised as a rule. This is an NGSS Science and Engineering Practice 4 (Analysing and Interpreting Data) applied to mathematics — learners analyse numerical data to identify relationships, rather than being handed a formula to memorise.	$\sqrt{(x'^2 + y'^2)} = \sqrt{(x^2 + y^2)}$ for each vertex? Teacher can spot-check by asking: 'Is A still the same distance from the origin in all three images?' (2) Are both partners recording — or is one passive? (3) In the pattern-spotting stage, are learners noticing the sign change and the swap? The critical success indicator is: at least one member of each pair can state a partial rule (e.g., 'the x and y seem to swap' for 90° ACW). Pairs who finish early are challenged: 'Can you write the pattern as an arrow rule using algebra, like $(x, y) \rightarrow (?, ?)$'
Explain Phase VIDEO: Rotating shapes Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Rotate points (basic) Source: Khan Academy (English - US curriculum) Search ARES for similar readings	The teacher leads a whole-class structured discussion to formalise the four coordinate rules. Learners copy the completed rules into their books and immediately apply them to two problems: (1) Rotate point P(3, -2) about the origin through 90° anticlockwise — calculate the image without drawing. (2) The Kasarani application: a mark on a football is at position Q(2, 5) relative to the ball's axle (treated as the origin). The ball rotates 90° clockwise then 180°. Calculate Q' and Q''. Learners then use the translate-apply-translate-back	Begin by pointing to the 'Patterns We Noticed' list on the board. Say: 'Let us now be mathematicians and turn these patterns into rules. I will ask three people to suggest the rule for 90° anticlockwise.' Wait 10 seconds. Cold-call three students. After each suggestion, test it publicly: 'Let us check: if we apply this rule to A(1, 1), do we get the image A' we constructed?' Once the correct rule $(x, y) \rightarrow (-y, x)$ is confirmed, write it prominently: 90° ACW about origin: $(x, y) \rightarrow (-y, x)$. Repeat this process for 90° CW — $(x, y) \rightarrow (y,$	Rule Codification from Inductive Evidence (Claim–Evidence–Reasoning): Learners first generated evidence (the coordinate tables), then identified patterns, and now formalise those patterns into algebraic claims. The teacher's deliberate strategy of testing each proposed rule against the already-constructed image (the evidence) models scientific reasoning — a rule is only accepted if it is consistent with all available evidence. The Kasarani context then immediately provides a novel application that checks	Teacher looks for: (1) Can learners correctly substitute into the rule without drawing a grid — this tests algebraic fluency with the rule, not just geometric intuition? (2) For the Kasarani problem, can learners apply the rule twice in sequence (composition of rotations)? The answer $Q(2, 5) \rightarrow 90^\circ \text{ CW} \rightarrow Q'(5, -2) \rightarrow 180^\circ \rightarrow Q''(-5, 2)$. (3) For the non-origin centre problem, can learners articulate the translate-apply-translate-back logic, or do they attempt to apply the rule directly to the original coordinates? This is the most common error —

	strategy to rotate a point about a non-origin centre: R(5, 3) rotated 90° anticlockwise about centre C(2, 1).	$-x$ — and for 180° — $(x, y) \rightarrow (-x, -y)$. After writing all three rules, say: 'Notice that 270° anticlockwise gives the same image as 90° clockwise. Can anyone explain why in one sentence?' Wait 15 seconds. Cold-call 2 students. Then say: 'Now apply your rules. Problem 1 — P(3, -2), rotate 90° anticlockwise. Work it out in your books. No grids. Just the rule. You have 90 seconds.' After 90 seconds cold-call one student to give the answer aloud and explain the substitution step by step. For the Kasarani problem, say: 'The marked spot on the ball is at Q(2, 5). The axle of the ball is the origin. The ball rolls forward, rotating 90° clockwise. Where is the mark now? Then it continues and rotates a further 180°. Where does the mark end up? Two steps — work them out.' Circulate and check working. For the non-origin centre problem, say: 'Our rules only work when the centre of rotation is the origin. But what if the centre is C(2, 1)? Discuss with your partner for 2 minutes: how could you shift the problem so that C becomes the origin?' After 2 minutes, cold-call one pair. Guide them to articulate: subtract the centre coordinates from the point, apply the rule, add the centre coordinates back. Work through R(5, 3) rotated 90° ACW about C(2, 1) step by step on the board: Step 1 — translate: $(5-2, 3-1) = (3, 2)$. Step 2 — apply rule: $(3, 2) \rightarrow (-2, 3)$. Step 3 — translate back: $(-2+2, 3+1) = (0, 4)$. So $R' = (0, 4)$.	whether the rule has been genuinely understood or merely copied.	flag it explicitly. Exit signal: teacher asks every learner to hold up their answer to Problem 1 written large on a mini-whiteboard or on a torn piece of paper — P(3, -2) rotated 90° ACW $\rightarrow$ P'(2, 3). Count how many are correct before moving on.
Driving Question Board (DQB)	Learners return to the class Driving Question Board displayed on the wall. In pairs, they write one	Point to the driving question displayed at the top of the DQB: 'When the minute hand of the	Public Knowledge Wall (Progressive Summarising): The DQB functions as a living	Teacher looks for: (1) Do learner cards reference the coordinate rules specifically, or do they

Creation  VIDEO: Rotating shapes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations (basic) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	new sticky note (or card) to add to the board. The note must complete one of two sentence starters — either: 'We can now answer part of the driving question because...' OR 'A new question we have is...'. Pairs share their cards aloud before posting them. The class then votes (thumbs up / thumbs sideways / thumbs down) on whether each new card moves the class closer to fully answering the driving question about the Nairobi clock tower.	clock at the University of Nairobi completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up — and how can we use those rules to rotate any shape to any position, on paper and on a grid?' Say: 'In Lessons 1 to 3 we answered part of this question — we know the properties, and we know how to construct the image. Today we went further. Talk to your partner for 90 seconds: what can we now say that we could not say before Lesson 4?' After 90 seconds, cold-call 4 pairs to share. As each pair reads their card, ask the class: 'Thumbs up if this moves us forward, sideways if it is a question still open, down if we disagree.' Post all cards. Highlight in a different colour any card that references the coordinate rule — say: 'This is significant — for the first time we can calculate, not just draw, where any point on the clock hand will be. That is a big step.' Add a teacher card to the board: 'Lesson 4 unlocked: coordinate rules for 90°, 180°, 270° rotations about the origin — and any centre.'	document of the class's growing understanding. By requiring learners to connect their new knowledge explicitly to the driving question — and by voting on whether each card advances the inquiry — the teacher ensures that each lesson's learning is contextualised within the bigger storyline rather than treated as an isolated topic. The colour-coded teacher card signals the epistemological leap made today (from geometric to algebraic).	remain vague ('we learned about rotation')? Specificity indicates depth of understanding. (2) Are any learner cards surfacing genuine new questions (e.g., 'What happens if we rotate by an angle that is not 90° or 180° on a grid?' or 'Can we rotate about a point outside the shape?')? These signal readiness for Lesson 5 (finding the centre and angle of rotation given object and image). (3) Do learners correctly identify what is still unanswered — e.g., how to find the centre of rotation given an object and its image? Record on the DQB for later lessons.
Model Building Phase  VIDEO: Rotating shapes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations (basic) Source: Khan Academy (English	Each learner opens their cumulative 'Rotation Model' booklet (a folded A4 sheet maintained across all 8 lessons) and adds a new panel for Lesson 4. In the panel they must include: (a) a labelled Cartesian grid showing triangle ABC and all three rotated images (90° ACW, 90° CW, 180°), colour-coded; (b) a rules box containing all four coordinate rules in arrow notation; (c) a worked example showing the Kasarani football calculation ($Q \rightarrow Q' \rightarrow Q''$) with each substitution	Say: 'Your Rotation Model booklet is the story of everything we are learning about rotation. Today's panel must show both the geometric picture AND the algebraic rules — because today we discovered that they are two ways of describing the same thing.' Distribute the booklets. Allow 5 minutes of quiet individual work. Circulate and check three things: (1) Are all four rules correctly written in arrow notation? (2) Does the Kasarani worked example show the substitution step	Cumulative Model Revision (Multimodal Representation): The model booklet requires learners to integrate geometric (visual), algebraic (symbolic), and narrative (verbal) representations of the same concept. Research in mathematics education shows that learners who can move fluidly between multiple representations develop more robust and transferable understanding than those who rely on a single representation. The extension task (clock face positions) explicitly	Teacher looks for: (1) Are all four coordinate rules correctly written — particularly the distinction between 90° ACW $(x, y) \rightarrow (-y, x)$ and 90° CW $(x, y) \rightarrow (y, -x)$? These are the most commonly confused. (2) Does the worked Kasarani example show both rotation steps in sequence, with the output of step 1 becoming the input of step 2? (3) Does the connection sentence demonstrate conceptual understanding — i.e., does it reference the fixed centre, preserved distance, or predictable

- US curriculum) Search ARES for similar readings	step shown; (d) a one-sentence connection: 'These rules explain the Nairobi clock tower phenomenon because...'. Learners who finish early add a fifth panel element: rotate the minute-hand tip (modelled as a point at (0, 5) on the clock face grid) through each multiple of 90° and list the four positions — connecting directly to 12:00, 12:15, 12:30, and 12:45 from the anchoring phenomenon.	explicitly — not just the answer? (3) Does the one-sentence connection mention either the fixed centre, the preserved distance, or the predictable new position? If a learner's connection sentence is vague (e.g., 'rotation happens on a grid'), prompt: 'Tell me how the rule $(x, y) \rightarrow (-y, x)$ connects to what the minute hand of the Nairobi clock is physically doing.' With 2 minutes remaining, cold-call two learners to read their connection sentences aloud. Respond to each with: 'What in our rules tells us that the distance from the centre is preserved?' Guide them to notice that the rule $(x, y) \rightarrow (-y, x)$ preserves $\sqrt{x^2 + y^2}$ because $\sqrt{((-y)^2 + x^2)} = \sqrt{x^2 + y^2}$. Write this on the board as a final insight.	closes the loop back to the anchoring phenomenon, reinforcing that the coordinate rules are not abstract exercises but descriptions of real physical motion.	position — or is it purely procedural ('I substituted into the rule')? Collect 5 booklets at random at the end of the lesson to review before Lesson 5, checking specifically for the accuracy of the rules box and the quality of the connection sentence.
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D. TEACHER REFLECTION

1. During the Observe Phase pattern-spotting, did most pairs independently identify at least one correct coordinate rule, or did the majority need the whole-class discussion to see the pattern? If fewer than half of pairs spotted a rule independently, consider providing a more scaffolded table (with the first row pre-filled) in Lesson 5's consolidation activity.

2. How many learners made the classic error of applying the origin-based coordinate rule directly to a point when the centre of rotation was NOT the origin (the Kasarani non-origin problem)? If more than one-third made this error, plan a 5-minute warm-up at the start of Lesson 5 that revisits the translate-apply-translate-back strategy with a new example.

3. Were the DQB cards learners produced specific and mathematically precise — referencing the coordinate rules by name — or were they vague and general? If cards were vague, consider introducing a DQB sentence frame in Lesson 5: 'The coordinate rule for [rotation type] is [rule], which means that [physical interpretation].'

4. Did the extension task (rotating the clock minute hand tip through multiples of 90° to recover the 12:00, 12:15, 12:30, 12:45 positions) successfully re-anchor the algebraic work in the phenomenon, or did it feel disconnected? Consider making this extension task the opening activity of Lesson 5 as a whole-class warm-up to strengthen the phenomenon connection.

5. In the Model Building Phase, did learners' connection sentences demonstrate conceptual understanding (referencing fixed centre, preserved distance, or predictable position) or purely procedural recall? Review the 5 collected booklets before Lesson 5 and use specific sentences — anonymised — as discussion prompts to distinguish procedural from conceptual understanding.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners plotted triangle ABC on a Cartesian grid and constructed its rotated images through 90° anticlockwise, 90° clockwise, and 180° about the origin using their geometrical sets. They recorded the before-and-after coordinates of all three vertices in a structured table and circled numerical patterns between the original and image coordinates. They also observed that the distance of each vertex from the origin is identical in the object and all three images, confirming the distance-preservation property of rotation on a coordinate grid.
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What did I learn?	From the coordinate tables, learners inductively discovered and then formalised four algebraic rules: 90° anticlockwise about the origin maps $(x, y) \rightarrow (-y, x)$; 90° clockwise maps $(x, y) \rightarrow (y, -x)$; 180° maps $(x, y) \rightarrow (-x, -y)$; and 270° anticlockwise gives the same result as 90° clockwise. They also learned the translate-apply-translate-back strategy for rotating about a centre other than the origin, and applied all rules to the Kasarani football context to calculate the new position of a marked point after successive rotations.
How does this explain the phenomenon?	The coordinate rules explain precisely what happens to every point on the University of Nairobi clock minute hand as it rotates: each point's distance from the centre (the clock's pivot) is preserved — confirmed algebraically because $\sqrt{(-y)^2 + x^2} = \sqrt{x^2 + y^2}$ — and its new position after any multiple of 90° can be calculated exactly without drawing. The minute hand tip at (0, 5) on a grid maps to (−5, 0) after 90° anticlockwise (12:15 position), (0, −5) after 180° (12:30), and (5, 0) after 270° anticlockwise (12:45), precisely matching the four clock positions from the anchoring phenomenon. Mathematical rules, not just construction, now govern where every point ends up.

LESSON 5 (80 minutes): INVESTIGATE: Finding the Centre and Angle of Rotation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	By the end of the lesson, learners will be able to determine the centre and angle of rotation given an object and its image, using perpendicular bisector construction and a protractor, on both a plain surface and the Cartesian plane.
Knowledge	Learners will know that: (i) the centre of rotation is equidistant from each object point and its corresponding image point; (ii) the perpendicular bisector of the segment joining any object point to its image point passes through the centre of rotation; (iii) the angle of rotation is the angle subtended at the centre of rotation between an object point and its corresponding image point; (iv) these two pieces of information — centre and angle — fully define a rotation and allow us to verify it by re-performing the rotation.
Skills	Learners will be able to: (i) construct perpendicular bisectors of two or more line segments joining corresponding object–image point pairs; (ii) identify the centre of rotation as the intersection of those perpendicular bisectors; (iii) measure the angle of rotation using a protractor; (iv) verify the result by rotating the original object using the found centre and angle and confirming it maps onto the image; (v) carry out the full procedure on both plain paper and the Cartesian plane.
Attitudes	Learners will appreciate that mathematics allows us to 'work backwards' — to reconstruct the hidden rule of a transformation from evidence alone — and will develop persistence and precision in geometric construction. Learners will respect each other's reasoning as they collaborate in groups to reach a shared conclusion.
Key Inquiry Question	How is rotation applied in real-life situations?
Purpose in Storyline	In Lessons 1–4 learners established the properties of rotation, performed rotations given a centre and angle, and applied rotation rules on the Cartesian plane. In Lesson 5 the investigation is reversed: learners are given only the object and its image — no centre, no angle — and must reconstruct the rotation from the geometric evidence. This mirrors the detective work that an engineer or architect does when analysing an existing rotated structure. The lesson advances the driving question by showing that the relationship between a shape and its rotated image contains enough information to recover every parameter of the rotation uniquely.
Safety Notes	Compasses have sharp points — learners must keep compass tips pointed downward when not in use and must not wave compasses in the air. Rulers and set-squares have sharp edges; learners must place them flat on the desk when passing them to a group member. No safety concerns beyond standard geometric-instrument precautions.

B. LESSON OVERVIEW

Learners work in groups of four on a structured investigation using plain paper and the Cartesian plane. Each group receives a printed sheet showing a triangle (or quadrilateral) and its image after an unknown rotation — no centre or angle is marked. Using only a ruler, compass, and protractor, learners construct perpendicular bisectors of two object–image point-pair segments, locate their intersection as the centre of rotation, and then measure the angle of rotation with a protractor. They verify their answer by performing the rotation using the found centre and angle and checking that it maps the object exactly onto the image. The entire investigation is anchored to the phenomenon: the clock hand at the University of Nairobi moved to a new position — today learners discover how to reverse-engineer exactly where the pivot is and how far it turned, using nothing but the hand's starting and ending positions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Geometric constructions: perpendicular bisector Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners are shown a large printed image (or projected slide) of the University of Nairobi clock tower with the minute hand drawn in two positions: pointing to 12 (straight up) and pointing to 4 (i.e., 120° clockwise). The centre of the clock face is covered with a circular sticker so learners cannot see it. Alongside the clock image is a simple geometric diagram: triangle ABC and its image triangle A'B'C' on plain paper — no centre marked, no angle labelled. Learners individually write answers to two questions in their exercise books: (1) 'Where do you think the fixed point (centre of rotation) is for the clock hands — inside the clock face, on its edge, or somewhere else? Explain your reasoning.' (2) 'If you were given only triangle ABC and triangle A'B'C' and nothing else, describe in words ONE strategy you might use to find the centre of rotation.' After 4 minutes of silent individual writing, learners share their predictions with a shoulder partner for 2 minutes. The teacher then cold-calls 3 pairs to share — recording key predicted strategies on the board under the heading 'Our Predictions' without evaluating them yet.	Display the clock image and the triangle diagram simultaneously. Say: 'Look carefully at both images. The clock hand has moved — but its pivot is hidden. The triangle has moved — but we have not told you where the centre is. In your books, answer the two questions on the board. You have 4 minutes of silent thinking time — no talking yet.' Start a visible 4-minute timer. After the timer: 'Turn to your shoulder partner and share your ideas. You have 2 minutes.' After partner talk, cold-call exactly 3 pairs using a random name stick. For each response say: 'Interesting — I am writing that on our Predictions board. We will come back to test it.' Do NOT confirm or reject any prediction. Write verbatim key phrases from learners (e.g., 'the centre is in the middle somewhere', 'draw lines connecting matching corners', 'use a compass to find equal distances'). Wait 10–12 seconds after each cold-call before prompting if silence falls.	Think-Write-Pair-Share with Anchored Prediction: By requiring individual written predictions before discussion, every learner commits to a personal hypothesis. Anchoring to the clock phenomenon activates prior knowledge about fixed centres from earlier lessons and creates cognitive dissonance (the centre is hidden — how do we find it?) that motivates the investigation. The public Predictions board creates accountability — learners will revisit it at the end of the lesson.	Observable indicator: every learner has written at least one sentence for each question before partner talk begins. Teacher circulates during silent writing and scans books — look specifically for whether learners reference 'equal distance from object and image' (shows readiness) or offer only vague guesses (shows need for scaffolding during the investigation). Note 2–3 learners who wrote the word 'perpendicular' or 'bisector' — they will be strategically placed in groups to act as peer scaffolds.
Observe Phase  VIDEO: Determining rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners work in pre-assigned groups of four. Each group receives: (a) Investigation Sheet 1 — plain paper with triangle PQR and image triangle P'Q'R' printed clearly ($P \rightarrow P'$, $Q \rightarrow Q'$, $R \rightarrow R'$ are labelled corresponding vertices); (b) Investigation Sheet 2 — a Cartesian plane (grid from -8 to 8 on both axes) with quadrilateral	Distribute materials as groups settle. Read Step 1 aloud with the class: 'We are going to be mathematical detectives. The clock hand moved — we only know where it started and where it ended. Your job is to find the hidden pivot and the hidden angle using construction.' Set a visible 20-minute group work timer.	Structured Group Investigation with Verification Loop (NGSS Science and Engineering Practice 3 — Planning and Carrying Out Investigations; SEP 6 — Constructing Explanations): The four-step structure ensures learners do not just find an answer but test it. The verification step (Step 4) is critical — it transforms	Observable indicators during circulation: (1) Both perpendicular bisectors are constructed with visible compass arcs — not just drawn freehand with a ruler. (2) The two bisectors intersect at a single point (not parallel, not widely divergent — if they are, the group has made a construction error and needs redirection). (3)

 EXERCISE: Determine rotations (basic) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	ABCD and image A'B'C'D' plotted; (c) a geometrical set (ruler, compass, protractor, set-square). Groups follow a structured investigation guide with four steps printed on the sheet: STEP 1 — On Investigation Sheet 1, draw the segment PP'. Construct its perpendicular bisector using your compass and ruler. Label it ℓ_1. STEP 2 — Draw segment QQ'. Construct its perpendicular bisector. Label it ℓ_2. Mark the point where ℓ_1 and ℓ_2 intersect. Label it O — this is your candidate centre of rotation. STEP 3 — Draw rays OP and OP'. Use your protractor to measure angle POP'. Record the angle and the direction (clockwise or anticlockwise). STEP 4 — VERIFY: Using O as centre and your measured angle, rotate triangle PQR. Check that R maps to R'. If it does, your centre and angle are correct. On Investigation Sheet 2 (Cartesian plane), repeat Steps 1–4 for quadrilateral ABCD. After completing both sheets, each group prepares a 90-second verbal report: state the centre coordinates, the angle, the direction, and whether verification succeeded.	Circulate using a rotation pattern — visit each group at least twice. At the first visit, ask: 'Show me your perpendicular bisector construction for PP'. How did you check that your bisector is truly perpendicular?' Wait 12 seconds for response. At the second visit, ask: 'You have found O. Before you measure the angle, predict — do you expect angle POP' to equal angle QOQ'? Why?' Wait 10 seconds. If a group finishes Steps 1–3 early, prompt: 'Excellent. Now do the verification — rotate P using O as centre and your angle. Does it land exactly on P'? What does that tell us about our construction?' If a group is stuck on Step 2 (perpendicular bisector construction), ask: 'Where does a point that is equidistant from Q and Q' always lie? Draw that locus.' Avoid giving the answer — use Socratic prompting only. With 5 minutes remaining, call: 'Groups, you have 5 minutes — make sure your Cartesian plane sheet is complete and prepare your 90-second report.'	the activity from a construction exercise into a genuine scientific test of a hypothesis. Requiring the procedure on both plain paper AND the Cartesian plane builds the connection between geometric reasoning and coordinate reasoning, directly advancing the driving question about predicting positions on a grid.	The measured angle is consistent across two pairs of corresponding points (e.g., angle POP' $\approx$ angle QOQ') — this is the key check. (4) On the Cartesian plane, the centre coordinates are stated as an ordered pair. Flag any group where the two bisectors do not intersect within the page — this indicates a systematic error in constructing perpendicularity and requires immediate targeted intervention.
Explain Phase  VIDEO: Geometric constructions: congruent angles Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Justify constructions Source: Khan	Each group delivers their 90-second report to the class (teacher keeps time strictly). After all groups have reported, the teacher leads a whole-class consolidation using a large projected or drawn diagram. Learners copy a summary into their books under the heading: 'How to Find the Centre and Angle of Rotation.' The teacher then poses three deepening questions for learners to discuss in their groups for 3 minutes before a whole-class cold-	Time each group report strictly: 'Group 1 — 90 seconds starting now.' After all reports, say: 'Notice something remarkable — different groups, different triangles, but the same method worked every time. Why does that method always work?' Pause 12 seconds. Write the key theorem on the board verbatim as learners dictate it: 'The perpendicular bisector of the segment joining any object point to its corresponding image point under a rotation passes through	Gallery Report + Socratic Deepening + Consolidation Check (NGSS SEP 6 — Constructing Explanations; SEP 7 — Engaging in Argument from Evidence): The group reports give every learner an opportunity to hear the method articulated in peer language. The three deepening questions push beyond procedure into structural understanding — specifically, Q2 (Wanjiku's claim) and Q3 (180° case) require learners to test the boundaries of the method. The	Observable indicators: (1) During group reports — can the reporter state BOTH the centre (as coordinates or a described point) AND the direction of rotation? (2) For Q1 — does the learner articulate 'one line vs. a unique point'? (3) Consolidation Check — compass arcs visible on the perpendicular bisectors (not freehand lines). Teacher records on a class list: tick = arcs visible + correct centre; circle = correct centre but no arcs; cross =

Academy (English - US curriculum) Search ARES for similar readings	call discussion: Q1 — 'We constructed TWO perpendicular bisectors. Why two and not just one?' Q2 — 'Wanjiku says: I only need to connect ONE pair of corresponding points to find the centre. Is Wanjiku correct? Explain.' Q3 — 'If the angle of rotation is exactly 180°, what do you expect to happen when you construct the perpendicular bisectors?' Learners then individually complete a 'Consolidation Check' — a single problem on their mini-whiteboards (or in books): given triangle KLM and its image K'L'M' on a pre-drawn grid, find the centre and angle of rotation. They hold up their boards when done.	the centre of rotation.' Underline 'any' and ask: 'Why does it work for ANY pair, not just a special one?' Cold-call 2 learners. For the three deepening questions, give 3 minutes of group discussion then cold-call one learner per question — wait 10 seconds after asking before accepting a response. For Q1 expected answer: one bisector gives a line of possible centres; a second bisector narrows it to a unique point. For Q2: Wanjiku is wrong — one bisector gives a locus, not a unique point. For Q3: the perpendicular bisectors will all be concurrent and the 'angle' will measure 180°; the centre will be the midpoint of each PP' segment. For the Consolidation Check: circulate rapidly, note how many learners have the correct centre marked with arcs visible, and how many have only guessed without construction.	Consolidation Check provides immediate individual evidence of transfer.	incorrect centre. This informs grouping for Lesson 6.
Driving Question Board (DQB) Creation VIDEO: Geometric constructions: angle bisector Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Angles of Rotation in Standard Positions: Clock Angles (PLIX) Source: CK-12 Search ARES for similar readings	The class returns to the Driving Question Board (DQB) displayed prominently at the front. The teacher reads the current driving question aloud: 'When the minute hand of the clock at the University of Nairobi completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up — and how can we use those rules to rotate any shape to any position, on paper and on a grid?' Learners individually write one sticky note (or a card if sticky notes are unavailable) with a NEW understanding they can now add to the DQB — beginning with the sentence stem: 'We can now...' and one remaining question beginning with 'We still wonder...' Two volunteers read their 'We can	Point to the DQB and say: 'Every lesson, we add to this board. Today we did something different — we worked backwards. Write your I can now... and I still wonder... cards. You have 3 minutes of silent writing.' After 3 minutes: 'Who will share their I can now...?' Cold-call 2 volunteers. After each share, ask the class: 'Does everyone agree this belongs on the DQB? Thumbs up, thumbs sideways, or thumbs down.' Resolve any thumbs-sideways or thumbs-down by asking: 'What would you change about the wording?' Post the class Evidence Card with ceremony: 'This is a big moment — this card shows we can now reverse-engineer a rotation. That is a powerful mathematical skill.' Point back to the clock	Driving Question Board Update with Evidence Card (Science and Engineering Practice 8 — Obtaining, Evaluating and Communicating Information): The DQB is a cumulative public record of the class's growing understanding. Requiring learners to write both a 'can now' and a 'still wonder' card maintains epistemic humility — learners track not just what they know but what remains open. The Evidence Card physically documents the lesson's contribution to the overall storyline, making the learning progression visible.	Observable indicators: (1) 'We can now...' cards use mathematical language — look for the words 'perpendicular bisector', 'centre of rotation', 'angle of rotation', 'verify'. Cards that only say 'we can find the centre' without any method reference suggest surface-level understanding. (2) 'We still wonder...' cards reveal productive next questions (e.g., 'What if the object and image overlap?', 'Does it work for rotations of more than 180°?') versus confusion-based questions (e.g., 'Why do we draw two lines?') — the latter signals the need for re-teaching in the next lesson's warm-up.

	now...' cards aloud before posting. The teacher posts a class-wide 'Evidence Card' on the DQB reading: 'LESSON 5 EVIDENCE: Given only an object and its image, we can use perpendicular bisector construction to find the hidden centre of rotation and a protractor to find the hidden angle. We verified this works on both plain paper and the Cartesian plane.' The teacher explicitly connects back to the phenomenon: 'If someone photographed the Nairobi clock at 12:00 and again at 12:20, and covered the centre of the clock face — we now have the tools to find exactly where the pivot is, using only the two positions of the hand.'	image: 'The clock engineers at the University of Nairobi know exactly where the pivot of that minute hand is — but now, even if someone removed that information, you could find it again.' Remind learners: 'Keep your I still wonder... questions — we will answer some of them in Lessons 6 and 7.'		
Model Building Phase  VIDEO: Geometric constructions: parallel line Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners retrieve their cumulative Rotation Model — a diagram they have been building since Lesson 1 that represents everything they know about rotation. In Lessons 1–4, the model included: a centre point O, an object and its image, radius lines showing equal distances, and the angle of rotation marked. Today, learners add a new layer to the model: they draw a second version of their diagram where the centre O is NOT labelled, and they annotate it with the reverse-engineering method — showing the perpendicular bisectors of PP' and QQ' as dashed construction lines meeting at O, with the note 'O found by perpendicular bisector construction.' They also annotate the angle of rotation with: 'Angle measured with protractor at O between OP and OP'.' Below the diagram, learners write a one-sentence 'Model Update	Say: 'Open your cumulative Rotation Model from your folders. You have been building this since Lesson 1. Today you are adding the reverse-engineering layer.' Display a teacher version of the model on the board and narrate as you add: 'I am drawing the construction lines — these dashed arcs are the compass marks for the perpendicular bisector of PP'. Watch the technique.' Draw the arcs deliberately and slowly. 'Now I draw the bisector line. I repeat for QQ'. Where they cross — that is O.' Then: 'Now annotate your own model. Add the dashed construction lines and the one-sentence Model Update Statement. You have 7 minutes.' Circulate and stamp or initial completed models. With 2 minutes remaining: 'If you have finished, add the clock extension at the bottom — sketch the Nairobi clock at 12:00 and 12:20 and show how	Cumulative Model Revision (NGSS SEP 2 — Developing and Using Models): The cumulative model is the lesson's primary sensemaking artefact. By requiring learners to add the reverse-engineering layer — not replace it, but add to it — the model physically represents the growth of understanding across the unit. The model is also the learner's most powerful revision tool: it shows properties, forward rotation, coordinate rules, and now reverse-engineering, all on one page. Connecting the model back to the Nairobi clock at the end ensures that the phenomenon remains the anchor, not an isolated example.	Observable indicators: (1) Dashed construction arcs visible on the model — not just two lines drawn with a ruler. (2) The one-sentence Model Update Statement contains the word 'uniquely' or an equivalent ('only one', 'exactly one') — this signals understanding that the method gives a unique solution. (3) In the clock extension, the perpendicular bisector is drawn between the two positions of the minute hand tip — not between the centre and the tip. Collect 6–8 models at random for teacher review before Lesson 6 to identify any persistent misconceptions about the construction technique.

	Statement: 'A rotation is uniquely determined by its object and image — the centre and angle can always be recovered by perpendicular bisector construction and angle measurement.' Learners who finish early add a 'clock translation' — they sketch the Nairobi clock at 12:00 and 12:20 and mark on it the perpendicular bisectors of the tip of the minute hand in both positions, showing how the method applies to the original phenomenon.	you would apply this method to the clock.' Close by saying: 'Hold up your model. Look at how much it has grown since Lesson 1. Next lesson, we explore rotational symmetry — how many times a shape maps onto itself as it turns. Your model will grow again.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your responses in your professional journal: (1) CONSTRUCTION ACCURACY — Did learners draw genuine compass-arc perpendicular bisectors, or did most groups use a ruler to estimate the midpoint and draw a freehand perpendicular? If the latter dominated, plan a 5-minute construction drill at the start of Lesson 6. (2) VERIFICATION STEP — Did groups actually perform the verification (Step 4) or skip it? If most skipped it, consider making verification a graded component of the next investigation to signal its importance. (3) DQB QUALITY — Review the 'We still wonder...' cards collected. Do learners' open questions align with the content of Lessons 6 and 7 (rotational symmetry, congruence from rotation)? If many cards ask about the 180° case or overlapping objects, address these explicitly in the Lesson 6 warm-up. (4) EQUITY — Were the same 3–4 learners answering all cold-call questions? Review your name-stick records and identify learners who have not yet been cold-called in this unit — prioritise them in Lesson 6. (5) CARTESIAN PLANE TRANSFER — Did learners correctly state the centre of rotation as an ordered pair (x, y) on Investigation Sheet 2? If not, this gap will affect Lesson 6's work on rotational symmetry on the grid and must be closed with a targeted warm-up activity.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that a triangle and its rotated image — with no centre or angle given — contain enough geometric information to reconstruct the rotation completely. They saw, through their own compass-and-ruler construction, that the perpendicular bisectors of two object–image point-pair segments always meet at a single point: the centre of rotation. They also observed that the angle measured at that centre between any object point and its image point is the same for all corresponding pairs.
What did I learn?	Learners learned that: (i) the centre of rotation is equidistant from each object point and its corresponding image point, so it lies on the perpendicular bisector of every object–image segment; (ii) two perpendicular bisectors are sufficient to uniquely locate the centre; (iii) the angle of rotation is measured at the centre between the object ray and the image ray, using a protractor; (iv) the result can be verified by performing the rotation and checking it maps the object onto the image; (v) this reverse-engineering method works on both plain paper and the Cartesian plane.
How does this explain the phenomenon?	The mathematical explanation is: for any rotation, every point P and its image P' are equidistant from the centre O (since rotation preserves distances). Therefore O lies on the perpendicular bisector of PP' . This is true for every corresponding pair, so the perpendicular bisectors of PP' and QQ' both pass through O — their intersection is therefore uniquely O . The angle of rotation is $\angle POP'$, which equals $\angle QOQ'$ for all corresponding pairs. This is a consequence of the isometric (distance-preserving) nature of

	rotation.
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LESSON 6 (80 minutes): Rotational Symmetry of Plane Figures and Solids

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To determine the order of rotational symmetry of plane figures and to identify the axis and order of rotational symmetry in solids, connecting these ideas to the Nairobi clock tower phenomenon and to Kenyan cultural and environmental contexts.
Knowledge	Learners will know: (a) the definition of rotational symmetry and the meaning of 'order of rotational symmetry'; (b) that a plane figure has rotational symmetry of order n if it maps onto itself exactly n times in one full 360° rotation about its centre; (c) that every figure has at least order 1; (d) that 3-D solids possess an axis of rotational symmetry and an associated order; (e) the order of rotational symmetry for equilateral triangles (order 3), squares (order 4), and regular hexagons (order 6).
Skills	Learners will be able to: (a) use paper cutouts of regular polygons to locate the centre of rotation and determine the order of rotational symmetry; (b) identify and describe the axis of rotational symmetry in 3-D solids (pyramids, triangular prisms, cones, tetrahedrons); (c) determine the order of rotational symmetry for each solid about a stated axis; (d) record findings systematically in a class table on the Driving Question Board.
Attitudes	Learners will appreciate: (a) the elegance and prevalence of rotational symmetry in Kenyan cultural artefacts such as the Maasai kiondo basket and Rift Valley windmill blades; (b) the value of collaborative investigation and accurate recording; (c) Responsibility — taking care of learning resources (paper cutouts and solid models) collected from the immediate environment to establish their axis and order of rotational symmetry; (d) Respect — appreciating each other's observations as they work in groups.
Key Inquiry Question	How is rotation applied in real-life situations?
Purpose in Storyline	Lessons 1–5 established what rotation is, how to perform it on a plane surface and the Cartesian plane, and how to find the centre and angle of rotation. Lesson 6 deepens understanding by asking: how many times does a shape look identical to itself during one full rotation? This 'order of rotational symmetry' idea directly extends the clock-hand phenomenon — the minute hand of the Nairobi clock returns to an identical-looking position once every 360° , giving it order 1 as an asymmetric hand but revealing order 12 if we think of the full clock face with all its hour-markings. By exploring kiondo basket patterns (regular hexagons, order 6) and Rift Valley windmill blades (equilateral triangles, order 3), learners connect the abstract mathematical idea to authentic Kenyan objects, preparing them for Lesson 7 (congruence from rotation) and Lesson 8 (summative consolidation).
Safety Notes	Scissors used for paper cutouts must be handled with care and stored point-down in a container when not in use. Solid models (especially cones and pyramids) may have sharp apexes — learners should handle them gently. No chemical or electrical hazards in this lesson.

B. LESSON OVERVIEW

Learners work in small groups using paper cutouts of regular polygons (equilateral triangle, square, regular hexagon) to discover rotational symmetry by physically rotating each cutout about a pin-held centre point and counting how many times it maps onto its outline in one full 360° turn. They then examine or handle models of 3-D solids — a square pyramid, triangular prism, cone, and tetrahedron — and identify the axis and order of rotational symmetry for each. All findings are entered into a class summary table on the Driving Question Board (DQB). Throughout, the teacher anchors every finding back to the Nairobi clock tower phenomenon: the minute hand sweeps a full 360° and the clock face itself exhibits rotational symmetry. The lesson explicitly connects to the Maasai kiondo basket (hexagonal weave pattern, order 6) and Rift Valley windmill blades (equilateral-triangle arrangement, order 3). The lesson follows the five-phase ARES structure: Predict → Observe → Explain → DQB Update → Model Building.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Intro to rotational symmetry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners are shown three images on the board side by side: (1) a photograph of the Nairobi University clock face, (2) a photograph of a Maasai kiondo basket viewed from above showing its hexagonal weave, and (3) a photograph of a three-bladed Rift Valley windmill viewed from the front. Each learner independently writes answers to two prompt questions in their exercise books: (a) 'If you rotated this object about its centre by less than 360°, at how many positions would it look exactly the same as it did at the start? Write a number for each image.' (b) 'What is special about those positions — can you describe the angle between them?' After 3 minutes of silent individual writing, learners share predictions with a partner (Think-Pair). No feedback is given yet — predictions are held for comparison after the Observe phase.	Display the three images and read the two prompt questions aloud. Say: 'Do not guess randomly — look carefully at the shapes and patterns in each image. Think about what you already know about rotation from our previous lessons.' Allow 3 minutes of silent individual thinking (DO NOT circulate or hint). After 3 minutes say: 'Now share your prediction with the person next to you — 90 seconds only.' After sharing, cold-call exactly 3 pairs (select one strong, one average, one hesitant learner from previous lessons). For each, ask: 'What number did you predict for the kiondo basket — and why?' Record the three predicted numbers for the hexagon on the board under the heading 'OUR PREDICTIONS'. Do not affirm or correct yet. Close by saying: 'Hold those predictions — we are about to test them with our hands.'	Think-Pair-Share with Anchored Prediction Recording. By writing predictions before any instruction, learners activate prior knowledge of rotation (from Lessons 1–5) and create a personal cognitive stake in the outcome. Recording predictions publicly on the board creates a reference point that will generate productive cognitive conflict or confirmation during the Observe phase, deepening encoding of the concept.	Observable indicator: Every learner has written at least one numerical prediction per image in their exercise book before the pair-share begins. The teacher scans books during the cold-call phase. Red flag: if more than one-third of learners write '1' for all three images, the teacher notes this as a misconception to address explicitly in the Explain phase — it suggests learners are confusing 'the shape looks different when rotated' with 'the shape maps onto itself'.
Observe Phase  VIDEO: Recognizing common 3D shapes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Rotational symmetry of 2D shapes Source: Khan	Groups of 3–4 learners each receive: one A4 sheet with printed outlines of an equilateral triangle, a square, and a regular hexagon (large enough to cut out, minimum 8 cm side); scissors; a blank A4 sheet; a sharp pencil; and a pin or drawing pin. Step 1 (8 min): Each group cuts out the three polygons. They place the equilateral triangle cutout on top of its printed outline on the blank sheet and push the pin through the centre point (teacher has pre-marked the centres). One learner holds the pin still while another slowly rotates	Distribute materials before the bell (pre-cut centre marks on polygon outlines save time). Circulate continuously during Step 1. At the 4-minute mark, stop the class for 30 seconds and say: 'Make sure the PIN is exactly at the centre I have marked — if it is off-centre, your count will be wrong. Check now.' At the 8-minute mark, cold-call one group: 'Group 3 — what order did you get for the equilateral triangle? How many degrees between each identical position?' (Wait 10 seconds.) During Step 2, prompt groups handling the cone:	Hands-On Manipulative Rotation with Structured Data Recording. Physical rotation of paper cutouts gives learners proprioceptive and visual evidence of the moment a shape 'clicks' back into its outline. Recording angles at each coincidence (120°, 240°, 360° for the triangle) before computing the order builds the conceptual link: $\text{order} = 360^\circ \div \text{angle between coincidences}$. The class compilation table (DQB entry) transforms individual group findings into shared public knowledge, enabling cross-group	Observable indicator: Each group's written record shows three separate entries (triangle, square, hexagon) with both the angle values and the order number correctly completed before Step 2 begins. The teacher checks at least two groups' written records during circulation. Red flag: A group recording order 6 for the square has likely miscounted — teacher intervenes immediately: 'Show me the six positions on your square cutout — point to each one.' A group recording order 1 for the regular hexagon has likely not

Academy (English - US curriculum) Search ARES for similar readings	the cutout clockwise. The group counts every position at which the cutout perfectly covers the outline again, including the starting position at 360°. They record: 'Equilateral triangle — maps onto itself at ____°, ____°, ____° — ORDER = ____'. They repeat for the square and the regular hexagon. Step 2 (10 min): Each group receives or examines one of four solid models (square pyramid, triangular prism, cone, tetrahedron — rotated between groups). Learners hold the solid, identify a vertical axis (using a pencil pushed through or held alongside), slowly rotate it, and count how many times the solid looks identical in one full 360° turn. They record: 'Solid: ____ — Axis description: ____ — ORDER = ____'. Step 3 (7 min): Groups share results with the class; teacher compiles a single class table on the DQB with columns: Shape/Solid Number of sides or faces Angle between identical positions Order of rotational symmetry.	'Is every position around the vertical axis identical for a cone? What does that tell you about its order?' (Wait 15 seconds before accepting answers.) During Step 3 (class table compilation), call on a DIFFERENT group for each solid. After the table is complete, point to the kiondo basket image and say: 'Look at the hexagon row in our table. Does the kiondo basket pattern match what we found? Talk to your partner — 30 seconds.' Cold-call 2 pairs. Then point to the clock image: 'The minute hand itself — just the hand, not the clock face — what order of rotational symmetry does it have? Why?' (Wait 10 seconds.) Expected answer: order 1 — it only maps onto itself after a full 360°. Validate and annotate the clock image on the board.	comparison and pattern recognition.	placed the pin accurately at the centre — teacher demonstrates correct re-pinning.
Explain Phase  VIDEO: Intro to rotational symmetry Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Determine rotations (basic) Source: Khan Academy (English - US curriculum) Search ARES for similar readings	The teacher leads a whole-class structured discussion to formalise the concept. Learners copy a definition into their notes: 'A plane figure has rotational symmetry of order n about a centre point C if rotating it by $(360^\circ/n)$ maps it exactly onto itself. Every shape has at least order 1.' Learners then individually complete three short written tasks in their exercise books: Task A — 'A regular pentagon: predict its order without cutting a cutout. Justify your answer using the formula $\text{angle} = 360^\circ/n$.' Task B — 'A Rift Valley windmill has 3 identical blades equally spaced. State its order of	Write the formal definition on the board and read it aloud. Emphasise: 'The key phrase is MAPS EXACTLY ONTO ITSELF — not just looks similar, but every point coincides.' Write the formula on the board: $\text{ORDER } n = 360^\circ \div$ (smallest angle of rotation that maps the shape onto itself). Then say: 'Work on Tasks A, B and C silently for 6 minutes.' Set a visible timer. After 6 minutes: 'Compare with your partner — 2 minutes.' Then cold-call: for Task A, call on a learner who was hesitant in Phase 1: 'What order did you get for the regular pentagon, and what angle?' (Wait 10 seconds.	Concept Definition Formalisation followed by Near-Transfer Application Tasks. Moving from physical manipulation (Phase 2) to formal symbolic definition and then immediately to application tasks (pentagon, windmill, prism) follows the concrete–pictorial–abstract progression. The windmill task (Task B) is a deliberate near-transfer to the Rift Valley context, ensuring learners experience the concept as generalisable beyond the classroom polygons. Cold-call sequencing (hesitant → average → strong) ensures all ability levels engage publicly with formalised reasoning.	Observable indicator: All learners have written the formal definition verbatim and completed all three tasks before the debrief. For Task A, the correct answer (order 5, 72°) requires applying the formula — not physical manipulation. A learner who writes order 5 but cannot explain why has surface recall without conceptual understanding; the teacher notes this and will revisit during the Model Building phase. For Task C, a learner who says order 2 (instead of order 3) for the triangular prism is confusing the number of rectangular faces with the triangular face symmetry —

	rotational symmetry and the angle between each coincident position.' Task C — 'The axis of a triangular prism runs through the centres of the two triangular faces. State the order of rotational symmetry about this axis.' After 6 minutes, learners compare answers with a partner, then the teacher conducts a whole-class cold-call debrief.	Expected: order 5, angle 72°.) For Task B, call on a different learner: 'What does the windmill order tell us — could you predict where the blade will be after the wind turns it by 120°?' (Wait 12 seconds.) For Task C, call on a third learner: 'What is the order for the triangular prism about that axis — and why does it match the triangle cutout order from our experiment?' (Wait 10 seconds.) Explicitly connect back to the phenomenon: 'Return to the Nairobi clock. If the clock face had only 3 markings equally spaced — like a windmill — what would its order be? What about 12 markings like a real clock face?' Write responses on the board beside the clock image.		teacher clarifies on the spot by holding up the model and rotating it.
Driving Question Board (DQB) Creation  VIDEO: Intro to rotational symmetry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The class revisits the full Driving Question Board. The teacher reads the Driving Question aloud: 'When the minute hand of the clock at the University of Nairobi completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up — and how can we use those rules to rotate any shape to any position, on paper and on a grid?' Learners are asked to discuss in pairs: 'What NEW understanding from today's lesson can we add to our board? What new questions has today raised?' After 2 minutes of pair discussion, each group contributes one sticky note (or the teacher writes their contribution on the board) under two DQB columns: WHAT WE NOW KNOW and NEW QUESTIONS WE HAVE. The teacher then reads back all current WHAT WE NOW KNOW entries from Lessons 1–6 to show the cumulative	Point to the DQB and say: 'Let us update our board. We started this unit asking how the Nairobi clock hand works mathematically. What can we now say that we could NOT say before today?' Allow 2 minutes of pair discussion. Then cold-call one pair per group (aim for 4–5 contributions). As each contribution is stated, the teacher writes it in the WHAT WE NOW KNOW column using the learner's own words, then adds a brief mathematical annotation. Example: a learner says 'the hexagon fits into itself six times' — teacher writes this, then annotates: 'Order 6; angle = $360^\circ \div 6 = 60^\circ$.' For the NEW QUESTIONS column, prompt: 'Has anyone wondered — if we rotate a shape and it maps onto itself, is that shape congruent to its rotated image? We will explore that in Lesson 7.' Add this as a teacher-seeded question. Close by pointing to the clock: 'The full clock	Cumulative Public Knowledge Construction on the DQB. The DQB functions as a living, co-authored record of the class's growing understanding. Reading back all Lesson 1–6 entries aloud activates retrieval practice across the entire evidence sequence, strengthening long-term retention. The teacher-seeded question about congruence creates a forward-looking bridge to Lesson 7, demonstrating to learners that their investigation is not yet complete — sustaining curiosity and epistemic motivation.	Observable indicator: Every group contributes at least one substantive entry to the DQB (not merely 'we learned rotation'). The teacher listens for mathematical precision — entries must include the word 'order' or a specific angle. Red flag: if groups contribute only qualitative statements ('shapes can turn and look the same'), the teacher prompts: 'Can you say that using a number and a degree symbol?' before accepting the contribution.

	progression of the sequence.	face with 12 hour-marks has order ____?' Cold-call. (Expected: 12. Angle: 30°.) Write it on the board as the capstone DQB entry for Lesson 6.		
Model Building Phase VIDEO: Intro to rotational symmetry Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Each learner opens their cumulative rotation model (begun in Lesson 1 and revised in each subsequent lesson). This model is a annotated diagram showing the Nairobi clock hand on a coordinate grid, with accumulated annotations from previous lessons (centre of rotation, angle of rotation, direction, invariant distances). Today, learners add a new layer: they draw the regular hexagon (like a kiondo basket cross-section) centred at the origin of the same coordinate grid, label its six vertices, and annotate: (a) the order of rotational symmetry (6); (b) the angle between each coincident position (60°); (c) an arrow showing the 60° rotation that maps vertex A to vertex B. Below the grid, learners write one sentence connecting today's finding to the clock: 'The Nairobi clock face with 12 hour-marks has rotational symmetry of order 12; each rotation of 30° maps the clock face onto itself.' Finally, learners write a 'model limitation note': 'My model shows a 2-D cross-section — the kiondo basket is 3-D; its full rotational symmetry must account for its axis and depth.'	Distribute learners' cumulative model sheets (or direct them to their exercise books where the model has been built). Say: 'Add today's hexagon to your existing coordinate grid — place it at the origin with one vertex at (0, r) where r is the distance you choose. Label all six vertices A through F going clockwise.' Allow 5 minutes for drawing. Circulate and check that: (1) the hexagon is regular (all sides equal, all angles 120°), (2) the centre is at the origin, (3) the 60° rotation arrow is correctly drawn from one vertex to the next. After 5 minutes, say: 'Now write the two sentences — the clock sentence and the model limitation note. 4 minutes.' After 4 minutes, cold-call 2 learners to read their clock sentence aloud. For each, ask: 'Does the rest of the class agree? Thumbs up, thumbs sideways, or thumbs down — show me now.' Address any thumbs-sideways or thumbs-down immediately. Close by saying: 'In Lesson 7, we will ask — if rotation maps a shape onto a congruent image, what kind of congruence is it? Your model will grow one more time.'	Cumulative Model Revision as Visible Thinking. Adding the hexagon layer to a model already showing the clock hand, coordinate grid, and previous rotation annotations makes the conceptual connection between rotational symmetry and the transformation of rotation explicit and spatial. The 'model limitation note' introduces epistemic humility — a core scientific and mathematical habit of mind — by asking learners to articulate what their 2-D model cannot yet represent about the 3-D kiondo or Rift Valley windmill. This prepares learners for the 3-D axis discussion in solids (already observed in Phase 2) and previews the depth of the concept.	Observable indicator: Each learner's model shows a correctly drawn regular hexagon at the origin with all six vertices labelled, a 60° rotation arrow, the order-6 annotation, and both required sentences written in full. The teacher collects 6–8 models at random at the end of the lesson (without pre-announcing which learners) to check for the model limitation note — this is the highest-order thinking requirement of the lesson and the most reliable indicator of deep understanding. Red flag: a learner who draws an irregular hexagon (unequal sides) has not transferred the definition of 'regular polygon' — teacher marks this for individual follow-up before Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During Phase 2, did all groups successfully count the correct order for the regular hexagon (order 6)? If more than two groups recorded an incorrect value, the pin-at-centre technique needs a more structured scaffold — consider pre-drilling a small hole at the centre of each cutout before the next time this lesson is taught. (2) In Phase 3, Task C (triangular prism order about its axis), did learners spontaneously connect the solid's order to the

triangular face order from Phase 2? If not, the link between plane figures and the cross-sectional face of a solid needs a more explicit bridge — consider adding a 'cross-section flash card' activity. (3) Did hesitant learners (identified in Phase 1 predictions) successfully answer cold-call questions in Phase 3? If not, review seating and cold-call strategy — these learners may need a structured partner with stronger understanding during Phase 2 in future lessons. (4) Were the DQB contributions in Phase 4 mathematically precise (including order numbers and degree values), or mostly qualitative? Qualitative-only contributions suggest learners have not yet consolidated the formal definition — begin Lesson 7 with a 3-minute retrieval quiz on the order formula before introducing congruence.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners physically rotated paper cutouts of an equilateral triangle, a square, and a regular hexagon about a pin-held centre point and counted the number of times each shape mapped exactly onto its own outline in one full 360° rotation. They also examined and rotated models of a square pyramid, triangular prism, cone, and tetrahedron to identify each solid's axis and order of rotational symmetry. Results were compiled into a class table on the Driving Question Board.
What did I learn?	A plane figure has rotational symmetry of order n if it maps onto itself exactly n times in a full 360° rotation about its centre. The smallest angle of rotation that achieves this is $360^\circ/n$. The equilateral triangle has order 3 (angle 120°), the square has order 4 (angle 90°), and the regular hexagon has order 6 (angle 60°) — matching the hexagonal weave pattern of a Maasai kiondo basket. Three-dimensional solids have an axis of rotational symmetry; the triangular prism has order 3 about its longitudinal axis, the square pyramid has order 4 about its vertical axis, the cone has infinite order about its vertical axis, and the tetrahedron has order 3 about an axis through one vertex and the centre of the opposite face. Every shape or solid has at least order 1.
How does this explain the phenomenon?	The order of rotational symmetry reveals how many times a shape reproduces its own appearance during one complete rotation about a fixed centre or axis — exactly as the minute hand of the Nairobi University clock returns to an equivalent position. A regular polygon with n sides has rotational symmetry of order n because rotating it by $360^\circ/n$ maps every vertex to the next identical vertex position. This is a direct consequence of the equal side lengths and equal interior angles that define a regular polygon. The same principle applies to the cross-sectional face of a prism: a triangular prism's order-3 symmetry about its axis is determined by the equilateral triangle at each end. The kiondo basket's hexagonal pattern (order 6) and the Rift Valley windmill's three-blade arrangement (order 3) are real-world applications of this mathematical rule, and the full 12-mark Nairobi clock face is a real-world example of order-12 rotational symmetry with a 30° rotation angle between each coincident position.

LESSON 7 (80 minutes): Rotation and Congruence: Same Shape, Different Story

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	By the end of the sub-strand the learner should be able to deduce congruence from rotation and appreciate the application of rotation in real-life situations.
Knowledge	Learners will identify and distinguish between direct congruence (produced by translation) and opposite congruence (produced by rotation and reflection), and explain why a rotated image cannot be superimposed on its object without lifting it from the plane.
Skills	Learners will classify pairs of object-and-image cutouts from previous lessons as directly or oppositely congruent, construct logical arguments to justify their classification, and connect congruence type to the transformation that produced it.
Attitudes	Learners appreciate the precision required in geometric reasoning, respect each other's arguments during sorting and justification activities, and take responsibility for the accuracy of their classifications.
Key Inquiry Question	How is rotation applied in real-life situations?
Purpose in Storyline	In Lessons 1–6, learners built up the full toolkit for performing rotations: properties, centre-finding, Cartesian rules, and rotational symmetry. Now, in Lesson 7, they ask a deeper question — when the minute hand of Nairobi's clock rotates from 12:00 to 3:00, is the hand at 3:00 'the same' as the hand at 12:00? Yes — but in exactly what mathematical sense? Learners classify the congruence type produced by rotation and contrast it with the congruence from translation and reflection, completing the congruence landscape before the unit synthesis in Lesson 8.
Safety Notes	Scissors are used in the sorting activity; learners must handle them carefully and pass them handle-first. Ensure adequate table space so that cutouts are not knocked onto the floor.

B. LESSON OVERVIEW

Lesson 7 is the EXPLAIN phase of the evidence-gathering sequence. Learners bring forward the object-and-image cutout pairs they produced during rotation activities in Lessons 3–5. Through a structured sorting activity they physically attempt to superimpose each image on its object — sliding, flipping, and lifting — to discover that rotated images require lifting from the plane to match their objects, classifying them as oppositely congruent. They compare this with translation pairs (directly congruent — no lifting needed) and reflection pairs (also oppositely congruent — lifting needed). A three-column sorting chart on the board consolidates understanding. The lesson closes by returning to the Nairobi clock: the hour hand at 3:00 and at 9:00 are congruent to the hand at 12:00 — but are they directly or oppositely congruent? Learners argue, justify, and update the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Identifying transformations	Learners are shown two large printed outlines of the University of Nairobi clock's minute hand: one pointing to 12 (straight up) and one	Display the two clock-hand outlines prominently. Say verbatim: 'I am not asking you to calculate anything yet. I want your	Prediction-with-justification (Claim–Evidence–Reasoning primer): By committing a written prediction before the activity,	Observable indicator: Every learner has a written YES/NO plus a sentence in their book before pair-share begins. Listen for

Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Translations, Rotations, and Reflections (Flexbook) Source: CK-12 Search ARES for similar readings	pointing to 3 (pointing right). The teacher places both cut-outs on the board and asks: 'These two hands are congruent — I promise you. But I want you to predict, before we test anything: if I pick up the 3 o'clock hand and try to slide it, without flipping it, onto the 12 o'clock hand — will it match perfectly?' Learners write a YES/NO prediction with a one-sentence justification in their exercise books. They then share with a shoulder partner for 90 seconds before three pairs are cold-called to share predictions with the class. A quick show-of-hands tally is recorded on the board (YES vs NO).	gut feeling — scientist's instinct. Write it down, then tell your partner.' Allow 90 seconds of pair talk, then cold-call exactly 3 pairs using the class register (rows 1, 3, and 5). Record the tally visibly. Do NOT correct predictions — say: 'We will find out together. Hold that thought.' Wait time after each cold-call: 12 seconds minimum before prompting.	learners activate prior knowledge about congruence and create a cognitive anchor. The subsequent tally makes disagreement visible, generating productive tension that the Observe phase will resolve.	justifications that reference 'sliding' vs 'turning' — this reveals whether learners already sense the direction-of-congruence distinction. Flag learners who say 'it doesn't matter how you move it' for targeted questioning during Phase 2.
Observe Phase  VIDEO: Defining transformations Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Translations, Rotations, and Reflections Practice Source: CK-12 Search ARES for similar readings	Groups of 3 receive an envelope containing four pairs of paper cutouts labelled with letters: Pair A (triangle and its translation image from Lesson 3), Pair B (triangle and its rotation image from Lesson 4), Pair C (quadrilateral and its reflection image from Lesson 2 sub-strand revision), and Pair D (the two clock-hand outlines from the predict phase). For each pair, groups must attempt three actions in order and record results on a structured observation table: (1) SLIDE only — can you make the image land exactly on the object by sliding alone? (2) FLIP then slide — does flipping (turning the cutout over) and then sliding make them match? (3) LIFT — do you need to pick the cutout completely up off the table to make them match? Groups record YES/NO for each action per pair, then write one sentence describing what they notice about which transformations required lifting.	Distribute envelopes and the printed observation table. Say: 'You have exactly 12 minutes. Do the three actions in order — slide, flip-then-slide, lift. Record honestly what you see. If your group disagrees, write both answers and be ready to argue.' Circulate and narrate what you see without giving answers: 'Interesting — I see that group trying to slide Pair B. What are you noticing?' At the 10-minute mark, say: 'Two more minutes — make sure every row of your table is filled.' At 12 minutes, call time. Cold-call 4 groups (one per pair) to report their observation-table findings for that pair. Wait 10 seconds after each group reports before asking the class: 'Does any group disagree with that finding?'	Manipulative-based evidence gathering (physical superimposition test): The three-step protocol (slide → flip-then-slide → lift) is a structured scientific procedure. It isolates the variable — which type of motion is needed — and makes the abstract distinction between direct and opposite congruence tangible and memorable. Connecting Pairs A/B/C/D to transformations learners themselves performed in earlier lessons grounds the new concept in their own prior work.	Observable indicator: Completed observation table with YES/NO entries for all 12 cells (4 pairs × 3 actions). Listen for groups that correctly identify that Pair A (translation) can be matched by sliding alone, while Pairs B and C (rotation and reflection) require lifting. If groups claim Pair B can be matched by sliding, prompt: 'Show me — keep the cutout flat on the table and try.' This exposes the misconception kinesthetically.

Explain Phase  VIDEO: Working with matrices as transformations of the plane Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mapping shapes Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The teacher introduces the formal vocabulary: DIRECT CONGRUENCE (image matches object by sliding alone — no lifting, no flipping) and OPPOSITE CONGRUENCE (image matches object only after lifting it off the plane — flipping it over). Learners copy a three-column summary table into their books: Transformation Congruence Type Why Lifting Is (or Is Not) Needed. They complete the table for translation, rotation, and reflection using the evidence from Phase 2. Next, learners tackle a worked-example card: a scalene triangle PQR with vertices P(1,2), Q(4,2), R(4,5) and its rotation image P'(-2,1), Q'(-2,4), R'(-5,4) under 90° anticlockwise rotation about the origin. They must (a) verify the two triangles are congruent by measuring side lengths, (b) attempt to describe whether a slide alone could superimpose P'Q'R' onto PQR, and (c) state the congruence type with justification. A class discussion follows, anchored by the question: 'Why does rotation produce opposite congruence even though it does not involve a mirror?'	Introduce vocabulary with a gesture: hold up a flat hand and slide it (direct) vs pick it up, flip it over, place it down (opposite). Write both terms on the board with colour coding — DIRECT in blue, OPPOSITE in red. Say verbatim: 'In your table, I want a mathematical reason in column 3, not just the word yes or no. A reason like: because the orientation of the vertices changes.' Give 8 minutes for the table and worked example. Then cold-call 2 learners for part (a), 2 for part (b), and 1 for part (c) — 5 different learners. After part (c) answer, pose to the class: 'Hands down — think for 12 seconds. Why does rotation flip orientation even though there is no mirror?' After 12 seconds, take 3 volunteer responses before giving the teacher explanation: rotation reverses the cyclic order of vertices (clockwise becomes anticlockwise), which is the geometric signature of opposite congruence.	Concept-mapping through structured note table: The three-column table forces learners to connect three representations — the transformation name, the abstract congruence category, and the physical/geometric reason. The worked coordinate example bridges the physical superimposition test (Phase 2) to formal mathematical verification, advancing from concrete to abstract (Bruner's CPA progression).	Observable indicator: Column 3 of the table contains a geometric reason (not just 'yes/no'). In the worked example, learners should state that $PQ = P'Q'$, $QR = Q'R'$, $PR = P'R'$ (congruent sides) but that the cyclic order $P \rightarrow Q \rightarrow R$ is clockwise while $P' \rightarrow Q' \rightarrow R'$ is anticlockwise, confirming opposite congruence. Misconception to watch: learners who confuse 'opposite congruence' with 'not congruent' — address immediately with: 'Opposite congruence still means same shape, same size. Only the orientation in the plane differs.'
Driving Question Board (DQB) Creation  VIDEO: Congruent shapes & transformations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Each learner receives two sticky notes. On the YELLOW note they write one thing that surprised them today — something they predicted differently in Phase 1. On the GREEN note they write one updated answer to any question currently on the class Driving Question Board (DQB). The teacher then draws the class's attention to the DQB and asks: 'Earlier in our unit, someone posted the question: Are all rotated images identical to the original	Hand out sticky notes. Say: 'Yellow = your biggest surprise today. Green = your best answer to any DQB question. You have 3 minutes — write independently, no talking.' After 3 minutes, collect sticky notes and read 4–5 yellow notes aloud anonymously: 'Someone wrote: I thought rotation was the same as translation in terms of congruence type. That is a great surprise to record.' Post selected green notes on the DQB. Then pose the clock question	Driving Question Board curation (metacognitive anchoring): Returning to student-generated questions and publicly updating answers models scientific knowledge-building — understanding grows iteratively. The clock anchor ensures that abstract congruence vocabulary is immediately grounded in the lesson's phenomenon, preventing inert knowledge.	Observable indicator: Green sticky notes contain specific, accurate DQB updates using the terms 'direct congruence' or 'opposite congruence' correctly. The clock question response should specify 'opposite congruence' with a rotation justification. Collect all sticky notes — they form an exit-ticket data set for planning Lesson 8 consolidation emphasis.

 EXERCISE: Rigid transformations: preserved properties Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	shape? Can someone now give a precise, upgraded answer to that?' Learners discuss in pairs for 60 seconds, then one pair shares. The teacher models writing a class-consensus update on the DQB card: 'Yes — rotated images are congruent to their objects (same shape, same size, same distances from centre), but the congruence is OPPOSITE — the cyclic order of vertices reverses, so the image cannot be superimposed on the object without lifting it from the plane.' Finally, learners revisit the clock scenario: 'The hour hand at 12:00 and the hour hand at 3:00 — what type of congruence connects them?' Learners write their answer on the back of their GREEN sticky note.	verbatim: 'Hour hand at 12 and hour hand at 3 — direct or opposite congruence?' Wait 12 seconds. Take 4 responses via cold-call. Confirm: 'Opposite congruence — the hand has been rotated, not slid, so its orientation in the plane has changed relative to the original position.'		
Model Building Phase  VIDEO: Linear transformation examples: Rotations in R2 Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Identify transformations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners open to their cumulative unit model page (started in Lesson 1). They add a new layer: a 'Congruence Landscape' diagram. This is a branching diagram with 'Congruence' at the top, splitting into 'Direct Congruence' (left branch, coloured blue) and 'Opposite Congruence' (right branch, coloured red). Under Direct Congruence they write: Translation — slide only, no lifting, no orientation change. Under Opposite Congruence they write: Rotation — lifting required, cyclic vertex order reverses; and Reflection — lifting required, mirror flips orientation. They then add one real-life Kenyan example for each branch: for translation they write 'A matatu sliding along Thika Superhighway — its shape and orientation remain unchanged'; for rotation they write 'The minute hand of Nairobi University clock —	Project the blank branching diagram on the board. Say: 'Copy this structure first — branches, colours, labels. You have 7 minutes to complete it and add your three Kenyan examples. Your examples must be different from mine.' Circulate and check that both Rotation and Reflection appear under Opposite Congruence — this is the most common omission. At 6 minutes, cold-call 3 learners to share their Kenyan examples. After sharing, say verbatim: 'Notice that our model is now connecting three lessons into one picture: Lesson 2 reflection, Lessons 3–5 rotation, Lesson 3 translation. That is what a good model does — it unifies.' Close by pointing to the phenomenon image: 'Every time the minute hand of Nairobi's clock completes a rotation, it produces a new position that is oppositely	Cumulative model revision (progressive knowledge integration): Adding the Congruence Landscape to the unit model makes explicit the relationship between transformation type and congruence type — knowledge that was implicit across three prior lessons. The Kenyan examples requirement forces learners to transfer abstract categories to concrete, familiar contexts, deepening encoding and preparing them for real-life application questions in Lesson 8 and beyond.	Observable indicator: The branching diagram correctly places Translation under Direct and both Rotation and Reflection under Opposite Congruence. The arrow back to the phenomenon is labelled with the correct congruence type. Common error to flag: learners who place Rotation under Direct Congruence — address by returning to the superimposition test: 'Go back to Pair B in your envelope. Can you slide the rotation image onto the object without lifting? Try it again right now.'

	rotating from 12 to 3 changes orientation (opposite congruence); for reflection they write 'A dhow on Lake Victoria reflected in calm water — mirror image requires lifting to superimpose'. Finally, learners draw an arrow from the 'Rotation' node back to the anchoring phenomenon image at the top of their model page, labelling it: 'Opposite congruence — orientation reverses but shape and size are preserved.'	congruent to the start position. Our model now explains why.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During Phase 2, were learners genuinely uncertain about whether lifting was needed for the rotation pair, or did they arrive at the answer too quickly — if so, was it because the cutouts were too obviously asymmetric? Consider using a more symmetric shape (e.g., an isosceles triangle rotated 180°) in future iterations to preserve productive uncertainty. (2) Did learners successfully distinguish between 'opposite congruence' and 'not congruent'? How many sticky notes conflated the two? Plan a brief warm-up clarification at the start of Lesson 8 if more than 25% of green sticky notes show this misconception. (3) Were the Kenyan examples in the Model Building phase original and mathematically sound, or did learners simply copy the teacher's examples? If the latter, increase the independence requirement next time by giving examples only after learners have written their own. (4) Did the DQB update feel genuine or performative? If learners struggled to identify which DQB question to answer, it may signal that the board needs to be reorganised by theme (properties / procedures / real-life connections) before Lesson 8's synthesis session.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners physically tested four pairs of object-and-image cutouts (translation, rotation, reflection, and the Nairobi clock hand) using a three-step superimposition protocol — slide, flip-then-slide, lift — and recorded which pairs required lifting to match.
What did I learn?	Rotation produces OPPOSITE congruence: the rotated image is the same shape and size as the object, but its cyclic vertex order is reversed, so it cannot be superimposed on the object without lifting it off the plane. Translation produces DIRECT congruence (slide only). Reflection also produces opposite congruence. The hour hand of Nairobi's clock at different positions exemplifies opposite congruence through rotation.
How does this explain the phenomenon?	The reversal of cyclic vertex order under rotation — clockwise becomes anticlockwise — is the geometric reason that rotation produces opposite congruence, even though no mirror is involved. This unifies the congruence landscape: direct congruence preserves orientation; opposite congruence reverses it, regardless of whether the reversal is caused by a rotation or a reflection.

LESSON 8 (80 minutes): Model Building, Application & Assessment: "Putting It All Together"

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate and apply all rotation concepts — properties, coordinate rules, centre and angle of rotation, rotational symmetry, and congruence — in an integrated final performance task anchored in the Nairobi Clock Tower phenomenon.
Knowledge	Learners demonstrate knowledge of: (a) properties of rotation in different situations; (b) rotation rules on the Cartesian plane for 90° , 180° , and 270° turns (clockwise and anticlockwise); (c) methods to determine the centre and angle of rotation given an object and its image; (d) order of rotational symmetry of plane figures; (e) axis and order of rotational symmetry in solids; (f) congruence as deduced from rotation.
Skills	Learners are able to: (a) rotate an object given the centre and angle of rotation on a plane surface and on the Cartesian plane; (b) determine the centre and angle of rotation given an object and its image; (c) determine the order of rotational symmetry of given plane figures; (d) determine the axis and order of rotational symmetry of given solids; (e) deduce congruence from rotation; (f) articulate a full written answer to the driving question using mathematical language.
Attitudes	Learners appreciate the application of rotation in real-life situations in their local Kenyan environment and demonstrate responsibility in using learning resources and respect for peers' mathematical reasoning.
Key Inquiry Question	How is rotation applied in real-life situations?
Purpose in Storyline	This is the final lesson of the rotation unit. Learners return fully to the anchoring phenomenon — the clock on the Administration Block of the University of Nairobi — and use it as the context for an integrated performance task that draws on every SLO developed across Lessons 1–7. They calculate exact coordinate positions of the clock hand at 12:00, 12:15, 12:30, 12:45, and 1:00; complete a structured class activity covering all seven SLOs; and compare their Lesson 1 prediction model with their final revised model to articulate how their understanding has grown. They close by writing a complete answer to the driving question and identifying rotation in their own Nairobi or local environment.
Safety Notes	No hazardous materials in this lesson. Learners using geometrical sets (compasses, protractors, rulers) must handle sharp instruments with care and store them safely after use. Ensure adequate desk space so that compasses do not cause injury during construction tasks.

B. LESSON OVERVIEW

Lesson 8 is the capstone lesson of Sub-Strand 2.3: Rotation. It is structured as a five-phase integrated performance and model-building session. In the Predict phase, learners revisit their original Lesson 1 clock-hand sketches and write a pre-task prediction for the coordinate positions of the Nairobi clock hand at four time positions. In the Observe phase, they watch a brief replay of the anchoring phenomenon clip and then receive a structured coordinate grid showing the clock hand at 12:00, giving them the exact starting data. In the Explain phase, they work through a seven-part class activity covering all SLOs: rotating on the Cartesian plane using rules, determining centre and angle of rotation by construction, finding the order of rotational symmetry of plane figures, identifying the axis and order of rotational symmetry of a solid, and deducing congruence. In the DQB phase, each learner writes a full, evidence-based answer to the driving question. In the Model Building phase, learners place their Lesson 1 sketch beside their final annotated model and articulate — in writing and orally — what changed in their understanding. A real-life appreciation task closes the lesson: learners identify two examples of rotation in their school or home environment (e.g., a ceiling fan at Nairobi's City Hall or the rotating drum of a githeri-grinding mill) and describe the centre, angle, and order of symmetry.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Intro to rotational symmetry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners retrieve their Lesson 1 prediction sketches from their exercise books. Each learner individually writes a short prediction in response to the prompt on the board: 'The tip of the University of Nairobi clock's minute hand starts at coordinate (0, 5) at 12:00, with the centre of rotation at the origin (0, 0). Predict the exact coordinates of the tip at 12:15, 12:30, 12:45, and 1:00. Write down the rotation rule you will use for each position.' Learners work silently for 5 minutes, writing predictions in their exercise books. They then share their prediction with a partner using a Think-Pair-Share, comparing whether they used the same rotation rules and the same coordinates. Pairs note any disagreements on a sticky note to post on the DQB 'We still wonder' column.	Teacher displays the original anchoring phenomenon image — the four clock positions on the University of Nairobi Administration Block — on the board or screen. Teacher says: 'Before we do any calculation, I want you to go back to Lesson 1. Take out the sketch you drew in your very first lesson. Look at what you predicted then.' [Allow 60 seconds for retrieval.] Teacher then reads the prediction prompt aloud and writes it on the board in full, including the coordinates (0, 5) and centre (0, 0). Teacher says: 'You now have all the tools you did not have in Lesson 1. Make your best prediction — use the rotation rules we have built together over eight lessons.' [WAIT TIME: allow 5 minutes of silent individual work.] Teacher circulates, scanning predictions without correcting — noting on a clipboard which rules learners are attempting (e.g., 90° anticlockwise: $(x, y) \rightarrow (-y, x)$). After pair-share, teacher cold-calls 3 pairs to state their predicted coordinates for 12:15. Teacher records the three different answers on the board without evaluating them yet: 'Let's hold all three ideas — we will test them shortly.'	Retrieval Practice + Prediction Conflict: Asking learners to revisit their Lesson 1 sketch before applying new knowledge activates prior knowledge and creates a productive cognitive conflict — they can see how their thinking has changed. Recording multiple predictions on the board without evaluating them builds curiosity and motivation to resolve the conflict through evidence, mirroring how scientists form and test hypotheses.	Observable indicator: Learner's written prediction correctly identifies the rotation angle for each quarter-hour (90° clockwise per 15 minutes) and attempts to apply a coordinate rule. Teacher scans for whether learners are confusing clockwise with anticlockwise rules, or whether they are applying translation instead of rotation. Misconceptions noted on clipboard will be targeted in Phase 3.
Observe Phase  VIDEO: Determining rotations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Teacher distributes a pre-printed A4 activity sheet (or learners rule one in their exercise books following teacher's board diagram). The sheet shows a Cartesian plane with the clock face centred at the origin. The minute hand at 12:00 has its tip plotted at point A(0, 5). Three additional labelled	Teacher says: 'Watch the clip one more time — this is the same phenomenon we started with eight lessons ago. But now you have a full mathematical toolkit. As you watch, I want you to notice and note: what is invariant — what never changes — and what is the variable — what changes each	Anchoring Return + Structured Observation: Returning to the original phenomenon video/images at the final lesson closes the narrative loop of the unit and signals to learners that all their accumulated knowledge now applies to this one motivating context. Asking learners to	Observable indicator: During the group sharing of observations, learners should state that distance from the centre (length of the hand), shape of the hand, and size are invariant, while direction/angle changes. Teacher listens to group discussions and intervenes if learners say 'the hand flips'

 EXERCISE: Determine rotations (basic) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	points mark the shaft of the hand: B(0, 3) and C(0, 1), so learners are rotating a line segment (model of the hand) rather than just a single point. Learners are also shown a short replay (2 minutes) of the anchoring phenomenon clip — or the teacher uses a printed four-image sequence pinned to the board — showing the hand at 12:00, 12:15, 12:30, and 12:45. Learners observe and record two things in their books: (1) What stays the same across all four images? (2) What changes? They share observations with their table group (groups of 4).	time.' [Play clip or reveal images one by one. Allow 2 minutes.] Teacher then distributes activity sheets. Teacher says: 'On your sheet, the clock centre is the origin. The tip of the hand at 12:00 is at A(0, 5). Points B and C are also on the hand. This is your object. Your task in the next phase will be to rotate this hand to its 12:15, 12:30, 12:45, and 1:00 positions.' Teacher asks the whole class: 'What is the angle the minute hand turns through every 15 minutes?' [WAIT TIME: 10 seconds.] Cold-calls 2 learners. Expected answer: 90°. Teacher confirms: 'Yes — 90° clockwise. And what rule do we use for a 90° clockwise rotation about the origin?' [WAIT TIME: 12 seconds.] Cold-calls 1 learner. Expected answer: $(x, y) \rightarrow (y, -x)$. Teacher writes this on the board and leaves it visible for Phase 3.	distinguish 'what stays the same' from 'what changes' deepens their understanding of the mathematical concept of invariance under rotation — a key idea in transformation geometry.	(confusing with reflection) or 'the hand moves sideways' (confusing with translation). These are targeted for correction in Phase 3.
Explain Phase  VIDEO: Intro to rotational symmetry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Angles of Rotation in Standard Positions: Clock Angles (PLIX) Source: CK-12  Search ARES for similar readings	Learners work through a structured seven-part class activity, individually first, then checking in pairs. The seven parts map directly to the seven SLOs of the sub-strand: PART A — Rotate on the Cartesian plane: Using the coordinate rule for 90° clockwise rotation $(x, y) \rightarrow (y, -x)$, learners calculate the image of each of the three points A(0,5), B(0,3), C(0,1) at 12:15. They repeat for 12:30 [180°: $(x,y) \rightarrow (-x,-y)$], 12:45 [270° clockwise = 90° anticlockwise: $(x,y) \rightarrow (-y,x)$], and 1:00 [360°: hand returns to original]. They plot all positions on the Cartesian plane and connect corresponding image points to form the hand at	Teacher introduces the seven-part activity by saying: 'This activity is your opportunity to show everything you have learned across this entire unit. Each part connects back to our clock — our phenomenon. Work individually first, then compare with your partner.' Teacher sets a visible timer: Parts A–B: 20 minutes; Parts C–E: 10 minutes; Part F: 5 minutes; Part G: 8 minutes. During Part A, teacher circulates and looks for correct application of rules. After 8 minutes, teacher pauses the class and says: 'Cold check — without looking at your partner: what are the coordinates of A' at 12:15 if A is at (0,5) and we rotate 90° clockwise?' [WAIT	Integrated Performance Task with Graduated Complexity: The seven-part structure moves from procedural fluency (Part A: applying coordinate rules) through conceptual understanding (Parts B–E: constructing, classifying, deducing) to transfer (Part F: real-life application) and synthesis (Part G: written argumentation). This matches NGSS Science and Engineering Practice 6 (Constructing Explanations) and Practice 8 (Obtaining, Evaluating, and Communicating Information). Requiring learners to use discipline-specific vocabulary in Part G forces them to consolidate and communicate their conceptual understanding rather than simply executing procedures.	Multiple checkpoints: (1) Cold-call on Part A coordinates — teacher records how many of the 4 cold-called learners give the correct clockwise answer; target is 3/4 or better before moving on. (2) During Part B construction, teacher checks that perpendicular bisectors are drawn with compasses (not just estimated by eye) — a specific, observable indicator of procedural accuracy. (3) Part G written paragraphs are scanned by teacher in the final 3 minutes — teacher looks for presence of at least 3 of the 6 target vocabulary terms as a minimum acceptable response indicator. Learners who write fewer than 3 terms are flagged for follow-up.

	each time. They label each image clearly. PART B — Determine centre and angle of rotation: Learners are given a second pre-drawn diagram showing a triangle PQR and its image P'Q'R' on a Cartesian plane (a scenario based on a rotating matatu wheel spoke in Nairobi's CBD). Using perpendicular bisectors of PP' and QQ', they construct the centre of rotation. They then measure the angle of rotation using a protractor. PART C — Order of rotational symmetry (plane figure): Learners are given outlines of three plane figures: (i) a regular hexagon (like the cross-section of a pencil), (ii) a rectangle (like a door of a classroom at their school), and (iii) the letter 'S'. They determine the order of rotational symmetry of each. PART D — Axis and order of rotational symmetry (solid): Learners examine a labelled diagram of a triangular prism (like the shape of a tent used at Uhuru Gardens). They identify the axis of rotational symmetry and state the order. PART E — Congruence from rotation: Learners are shown four pairs of triangles on a grid. They must identify which pairs are congruent by rotation (not reflection or translation), stating the evidence: same side lengths, same angles, same orientation type (direct congruence).	TIME: 12 seconds.] Cold-calls 4 learners. Records answers on board. If any learner gives $(-5, 0)$ — the anticlockwise answer — teacher does NOT immediately correct but says: 'Two different answers are on the board. Turn to your partner: which is 90° clockwise and which is 90° anticlockwise? How do you know?' [Allow 2 minutes of pair discussion.] Then resolves with whole class: the rule $(x,y) \rightarrow (y,-x)$ gives clockwise; $(x,y) \rightarrow (-y,x)$ gives anticlockwise. For A(0,5): clockwise $\rightarrow (5,0)$. Teacher confirms: 'At 12:15, the tip of the Nairobi clock hand is at (5,0). Does that make sense looking at the clock image?' [Points to the 3 o'clock position on the diagram.] During Part B, teacher says: 'Remember — to find the centre of rotation, we draw the perpendicular bisector of the segment joining each original point to its image. The centre is where these bisectors meet. This is a construction — use your compasses and ruler carefully.' Teacher demonstrates the first perpendicular bisector on the board for PP' while learners watch, then says: 'Now you complete it for QQ'. The intersection is your centre.' For Part C, teacher cold-calls 3 learners for the order of symmetry of the regular hexagon. Expected: 6. For the rectangle: 2. For 'S': 2. Teacher asks: 'Why does the letter S have order 2 but not order 4?' [WAIT TIME: 10 seconds.] Cold-calls 1 learner.		
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	PART F — Real-life appreciation task: Learners identify TWO examples of rotation in their school or home environment in Nairobi or their locality. For each they write: (a) what is rotating, (b) the approximate centre of rotation, (c) the angle or range of rotation, and (d) the order of rotational symmetry if applicable. Examples might include: a ceiling fan at City Hall, the rotating beater blades of a uji-stirring machine at home, the wheels of a mkokoteni (handcart) in Gikomba market, or the hands of a wall clock in the school office. PART G — Written answer to the driving question: Learners write a minimum 5-sentence paragraph answering: 'When the minute hand of the clock at the University of Nairobi completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up — and how can we use those rules to rotate any shape to any position, on paper and on a grid?'	For Part G (written answer), teacher says: 'Your paragraph must include at least four of these mathematical terms — write them on your page as a checklist before you start: centre of rotation, angle of rotation, coordinate rule, invariant, congruence, rotational symmetry.' Teacher writes these six terms on the board.		
Driving Question Board (DQB) Creation  VIDEO: Intro to rotational symmetry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine rotations	Each learner reads back their Part G written paragraph silently. They then write their final answer to the driving question on a fresh sticky note (or in a designated box on the activity sheet) and post it on the DQB under the column 'We now know'. They also retrieve any sticky notes they posted in earlier lessons under 'We wonder' and check whether those questions have now been answered. As a table group, they do a 2-minute 'DQB sweep': identify any questions still unanswered and	Teacher directs learners' attention to the full DQB display (which has been built up across all 8 lessons). Teacher says: 'This board has been our running record of thinking since Lesson 1 — every question you asked, every idea you posted. Today we complete it.' Teacher reads aloud 2–3 of the most interesting 'We wonder' questions posted in earlier lessons — for example, 'Can you rotate about a point outside the shape?' or 'What happens if the angle is not a multiple of 90°?' Teacher says:	DQB Closure — Narrative Arc Completion: The DQB has functioned throughout the unit as a public record of the class's evolving understanding. Completing it in the final lesson provides learners with a visible, concrete representation of their intellectual growth. Reading aloud earlier questions and answering them in the final lesson demonstrates that science and mathematics are not a collection of facts but a progressive process of inquiry — directly mirroring NGSS	Observable indicator: Each learner's sticky note contains a complete, mathematically accurate answer to the driving question — specifically naming: (a) the centre of rotation as the fixed point, (b) a specific coordinate rule, (c) the angle corresponding to each clock position, and (d) the concept of congruence between object and image. Teacher reads 5–6 sticky notes as learners post them and selects 2 exemplary ones to read aloud as models, naming the learner (with permission) and

Source: Khan Academy (English - US curriculum) Search ARES for similar readings	decide whether to post them on a final 'Still wondering' column (which the teacher commits to addressing or directing learners to explore independently). One representative from each table group reads aloud one 'We now know' statement to the class.	'Have we answered these? Raise your hand if you can answer the first one.' [WAIT TIME: 10 seconds.] Cold-calls 1 learner. Teacher then says: 'Now write your final answer to the driving question on your sticky note. You have 3 minutes. Be precise — use the mathematical language from Part G.' [Allow 3 minutes silent writing.] After posting, teacher invites one representative from each table to read their 'We now know' statement. Teacher synthesises by pointing to the clock image and saying: 'From the very first lesson, this clock asked us a question. Today, you answered it — with coordinate rules, construction, symmetry, and congruence. That is what mathematicians do.'	Practice 1 (Asking Questions) and Practice 6 (Constructing Explanations). This also supports metacognitive development as learners reflect on what they did not know in Lesson 1 versus what they know now.	highlighting the mathematical precision used.
Model Building Phase  VIDEO: Intro to rotational symmetry Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Determine rotations Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners open their exercise books to their Lesson 1 sketch — the rough drawing they made of the clock hand in Lesson 1 when they first encountered the phenomenon. They place this beside their completed Phase 3 activity sheet (their final model). Working individually, they complete a 'Model Comparison Card' with three prompts: (1) 'In Lesson 1, I thought the clock hand moved because...'; (2) 'I now know that the mathematical rule governing every point on the hand is...'; (3) 'The biggest idea I added to my model between Lesson 1 and today is...'. Learners then share their Model Comparison Cards with a partner. Finally, the class does a whole-group share: teacher asks 3 volunteers to read their response to prompt (3) aloud. The lesson closes with the teacher pointing to both models side by	Teacher says: 'Take out your Lesson 1 sketch right now — the very first drawing you made of the Nairobi clock hand. Place it on the left side of your desk. Place your activity sheet from today on the right.' [Allow 30 seconds for retrieval.] Teacher says: 'Look at both. In silence, I want you to think: what did your Lesson 1 you know? What does your Lesson 8 you know that Lesson 1 you did not?' [WAIT TIME: 15 seconds.] Teacher then distributes or dictates the three Model Comparison Card prompts, writing them on the board. Learners write individually for 5 minutes. Teacher circulates and reads over shoulders — noting particularly strong responses to prompt (3) for whole-class sharing. After pair-share, teacher cold-calls 3 volunteers (including at least one learner who was identified as	Model Comparison (L1 vs Final) + Transfer to Real World: Asking learners to physically place their Lesson 1 sketch beside their final model makes the growth in their understanding tangible and visible. This is a metacognitive sensemaking strategy — learners are not just recalling facts but evaluating the quality and completeness of their own prior and current models. The real-life appreciation task extends the sensemaking beyond the classroom into Nairobi and the learners' local communities, completing the transfer cycle from phenomenon → abstraction → real-world application. This directly addresses SLO (g): 'appreciate the application of rotation in real-life situations' and the KICD value of Responsibility.	Observable indicator: Model Comparison Card prompt (2) — 'I now know that the mathematical rule governing every point on the hand is...' — must contain a specific coordinate rule (e.g., 90° clockwise: $(x,y) \rightarrow (y,-x)$) to be counted as meeting expectations. Teacher collects 5 randomly selected cards at the end of the lesson to check against this indicator. Additionally, teacher observes during the pair-share whether learners can articulate the difference between their L1 model (descriptive, qualitative) and their L8 model (precise, rule-governed, quantitative) — this indicates whether the sensemaking across the full unit has been successful.

	side and saying: 'This is what learning looks like in mathematics — a model that started with a sketch and grew into a precise, rule-governed, evidence-based explanation.' As a take-home task (real-life appreciation), learners are asked to identify two examples of rotation in their school or home environment in Nairobi or their locality, and for each write: what is rotating, the centre of rotation, the approximate angle of rotation, and the order of rotational symmetry.	struggling in earlier lessons) to read their response to prompt (3). Teacher responds to each with a specific, content-focused affirmation: e.g., 'You mentioned coordinate rules — can you tell us which specific rule you added to your model?' [WAIT TIME: 10 seconds.] Teacher closes the lesson by saying: 'For your take-home task, I want you to be mathematicians in your community. Find two rotating things in your school or at home — it might be the school gate swinging open, the blades of your jiko fan, the wheel of a mkokoteni in the market, or the hands of the clock in the school office. Describe each one using the four vocabulary terms: centre of rotation, angle of rotation, order of rotational symmetry, and congruence. Write it up in your exercise book and bring it to the next lesson.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) When learners compared their Lesson 1 sketches with their final models, were they able to articulate specific conceptual additions (e.g., coordinate rules, construction methods, symmetry order) rather than just saying 'I know more now'? If not, how might future lessons better scaffold metacognitive language throughout the unit rather than only at the end? (2) During Part A of the class activity, how many learners applied the correct clockwise rule $(x,y) \rightarrow (y,-x)$ without prompting? If fewer than 70% of the class did so correctly, consider incorporating a quick daily rotation-rule drill in the opening 3 minutes of future geometry lessons. (3) Did learners' written answers to the driving question (Part G and the DQB sticky note) demonstrate genuine synthesis — connecting all four pillars of rotation (fixed centre, measurable angle, preserved shape/size, predictable new position) — or did they describe only one or two ideas? Use this to inform the framing of the driving question for future cohorts. (4) Were the real-life examples learners identified in the take-home task drawn from their genuine local Nairobi/Kenyan context (ceiling fans, mkokoteni wheels, jiko components, school gate hinges) or did they default to Western textbook examples? Kenyan contextual fluency in mathematical thinking is a goal of CBE and should be celebrated and reinforced when present. (5) Consider whether a brief peer-assessment of the Model Comparison Cards using a simple 3-point rubric (Developing / Meeting / Exceeding) in the following lesson would deepen learners' understanding of what a strong final explanation looks like in mathematics.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners re-watched the anchoring phenomenon — the University of Nairobi Administration Block clock — and observed the
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	minute hand (modelled as a line segment with points A(0,5), B(0,3), C(0,1) on a coordinate grid centred at the origin) at 12:00, 12:15, 12:30, 12:45, and 1:00. They observed that the length of the hand, the shape, and the distance of every point from the centre remained invariant at all four positions, while the direction of the hand changed by exactly 90° clockwise every 15 minutes.
What did I learn?	Learners learned and applied all seven SLOs of Sub-Strand 2.3: (a) rotation on the Cartesian plane using rules — 90° CW: $(x,y) \rightarrow (y,-x)$; 180°: $(x,y) \rightarrow (-x,-y)$; 270° CW / 90° ACW: $(x,y) \rightarrow (-y,x)$; 360°: $(x,y) \rightarrow (x,y)$; (b) construction of the centre of rotation using perpendicular bisectors of segments joining object points to their images; (c) order of rotational symmetry of plane figures (regular hexagon = 6, rectangle = 2, letter S = 2); (d) axis and order of rotational symmetry of a triangular prism solid; (e) deduction of congruence (direct congruence) from rotation; (f) real-life identification of rotation in Nairobi and Kenyan local contexts; (g) synthesis of all concepts into a full written answer to the driving question.
How does this explain the phenomenon?	The minute hand of the University of Nairobi clock is a real-world example of rotation: a geometric transformation in which every point of an object turns through a fixed angle about a fixed centre (the origin/clock centre), preserving all distances and angles. The mathematical rules that govern exactly where every point ends up are coordinate rotation rules: for a 90° clockwise rotation, $(x,y) \rightarrow (y,-x)$; for 180°, $(x,y) \rightarrow (-x,-y)$; for 270° clockwise, $(x,y) \rightarrow (-y,x)$. These rules allow us to predict the precise location of every point on any shape after any rotation on the Cartesian plane. The object and its image are always directly congruent (same shape, same size, same orientation type). The clock hand itself has rotational symmetry of order 1 (as a non-symmetric shape), while the circular clock face has infinite rotational symmetry — both are explained by the same underlying mathematical structure of rotation.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.